# SIMULATION IDEAS

The model is assumed to be

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i)\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \approx \sum_{s=1}^{S}\sum_{r=1}^{R} x^*_{ijrs} \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t_i)\right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} = \left\langle \mathbf{X}^*_{ij}, \mathbf{A}(t_i)\right\rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij} \tag{1}$$

where matrix $\mathbf{X}^*_{ij} := (x^*_{ijrs}) \in \mathbb{R}^{R\times S}$, and $\mathbf{A}(t) := (\alpha_{rs}(t)) \in \mathbb{R}^{R\times S}$ with $\alpha_{rs}(t) := \left\langle \mathcal{U}_r^{(s)}, \mathcal{A}^{(s)}(t)\right\rangle$. Within one patient, the eyes are correlated by

$$\begin{bmatrix}\epsilon_{i1}\\ \epsilon_{i2}\end{bmatrix} \sim \mathbf{N}\left(\mathbf{0},\ \sigma^2(t)\begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\right).$$

## Case 1: 2D Tensor Covariate

**DGP.** The tensor size is set to be $D = 2$, $p_1 = p_2 = 60$, and split by $p_1' = p_2' = 12$ such that by $S^{(1)} = S^{(2)} = 5$. Set the rank $R = 4$, and the $s$-th block tensor for $j$-th eye of $i$-th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^{R} x^*_{ijrs}\mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}, \quad \text{where } x^*_{ijrs} \overset{\text{iid}}{\sim} \mathrm{N}(0,1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^{R}\alpha_{rs}(t)\mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)}$ where $\alpha_{rs}(t)$ has 4 choices:

$$\begin{aligned}
&\alpha_{1rs}(t) = 5\sqrt{\frac{rs}{RS}}(t-0.2)^2, \quad \alpha_{2rs}(t) = \sqrt{\frac{rs}{RS}}\left(e^{-(3t-1)^2} - 0.75\right), \quad \alpha_{3rs}(t) = \sqrt{\frac{rs}{RS}}\sin\big(2\pi(t-0.5)\big),\\
&\alpha_{4rs}(t) = \sqrt{\frac{rs}{RS}}\Big[0.45\cdot 5(t-0.2)^2 + 0.35\big(e^{-(3t-1)^2} - 0.75\big) + 0.20\sin\big(2\pi(t-0.5)\big)\Big],\\
&\alpha_{5rs}(t) = \sqrt{\frac{rs}{RS}}\left[1.10\exp\left\{-\frac{(t-0.30)^2}{2(0.08)^2}\right\} - 0.95\exp\left\{-\frac{(t-0.72)^2}{2(0.11)^2}\right\}\right],\\
&\alpha_{6rs}(t) = \sqrt{\frac{rs}{RS}}\Big[18(t-0.2)(t-0.55)(t-0.85) - 0.05\Big]
\end{aligned}$$

The one-way vector $\mathbf{Z} \in \mathbb{R}^{n\times p_0}$ is Gaussian ensemble matrix (iid entries) and $\boldsymbol{\beta} = (2.0, 1.0, -1.0, 0.5)^\top$. We will vary $\sigma^2(t)$ to 4 subcases as Case 1(a)-(d).

$$\sigma_0^2(t) \equiv 1, \quad \sigma_1^2(t) = 1 + 0.3\sin(2\pi t), \quad \sigma_2^2(t) = 1 + 0.25\cos(2\pi t) + 0.1\sin(4\pi t), \text{ and } \sigma_3^2(t) = 0.5 + 0.5\sin^2(\pi t). \tag{2}$$

**Estimation Steps.** (1) We treat the paired data as iid, and get initial estimate $\hat{\mathbf{A}}^\dagger(t)$ and $\hat{\boldsymbol{\beta}}^\dagger$. (2) Then estimate covariance structure $\hat{\sigma}^2(t)$ by eq(15)-(16) in AOAS draft. (3) Get final estimates $\hat{\mathbf{A}}^*(t)$ and $\hat{\boldsymbol{\beta}}^*$ from eq(14)-(15) of AOAS draft.

**Simulation Results.** We vary $n = 1000/2000$, coefficients $\alpha_{1rs}(t), \ldots, \alpha_{4rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. We report the error of both initial estimates $(\hat{\mathbf{A}}^\dagger,\ \hat{\boldsymbol{\beta}}^\dagger)$ and final estimates $(\hat{\mathbf{A}}^*,\ \hat{\boldsymbol{\beta}}^*)$, also the error of $(\hat{\sigma}^2, \hat{\rho})$.

Table 1: MAE errors ($\times 10^{-2}$) for Case 1(a) - $\sigma_0(t) \equiv 1$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 1000 | 0.0 | **5.06 (± 0.17)** | 5.08 (± 0.18) | **2.09 (± 0.78)** | 2.12 (± 0.77) | 30.32 (± 2.66) | 2.81 (± 1.91) |
| | | 0.3 | 5.05 (± 0.20) | **4.94 (± 0.18)** | **2.44 (± 1.01)** | 2.49 (± 1.07) | 30.23 (± 3.48) | 7.48 (± 3.06) |
| | | 0.6 | 5.06 (± 0.24) | **4.53 (± 0.17)** | **2.66 (± 1.10)** | 2.69 (± 1.13) | 30.19 (± 3.71) | 15.09 (± 2.55) |
| | | 0.9 | 5.06 (± 0.27) | **3.57 (± 0.18)** | 2.88 (± 1.13) | **2.84 (± 1.15)** | 30.14 (± 3.99) | 22.70 (± 1.92) |
| | 2000 | 0.0 | **3.29 (± 0.10)** | 3.30 (± 0.10) | 1.25 (± 0.43) | **1.24 (± 0.44)** | 13.31 (± 1.88) | 1.37 (± 0.80) |
| | | 0.3 | 3.32 (± 0.11) | **3.22 (± 0.11)** | 1.39 (± 0.59) | **1.36 (± 0.61)** | 13.24 (± 1.82) | 4.12 (± 1.64) |
| | | 0.6 | 3.33 (± 0.11) | **2.89 (± 0.10)** | 1.53 (± 0.65) | **1.47 (± 0.67)** | 13.15 (± 2.04) | 8.48 (± 1.41) |
| | | 0.9 | 3.33 (± 0.11) | **2.16 (± 0.07)** | 1.68 (± 0.70) | **1.56 (± 0.72)** | 13.07 (± 2.32) | 12.92 (± 1.03) |
| $\mathbf{A}_2(t)$ | 1000 | 0.0 | **5.08 (± 0.21)** | 5.10 (± 0.21) | **2.09 (± 0.79)** | 2.12 (± 0.78) | 30.33 (± 2.72) | 2.94 (± 2.11) |
| | | 0.3 | 5.03 (± 0.18) | **4.92 (± 0.16)** | **2.44 (± 0.98)** | 2.51 (± 1.03) | 30.28 (± 3.56) | 6.87 (± 2.92) |
| | | 0.6 | 5.03 (± 0.20) | **4.50 (± 0.16)** | **2.66 (± 1.08)** | 2.71 (± 1.11) | 30.25 (± 3.78) | 14.46 (± 2.31) |
| | | 0.9 | 5.02 (± 0.22) | **3.52 (± 0.15)** | 2.89 (± 1.12) | **2.84 (± 1.14)** | 30.20 (± 4.05) | 22.09 (± 1.48) |
| | 2000 | 0.0 | **3.28 (± 0.08)** | 3.29 (± 0.08) | **1.21 (± 0.46)** | 1.21 (± 0.46) | 13.43 (± 1.89) | 1.56 (± 0.96) |
| | | 0.3 | 3.29 (± 0.12) | **3.19 (± 0.11)** | 1.39 (± 0.59) | **1.37 (± 0.63)** | 13.34 (± 1.78) | 3.87 (± 1.59) |
| | | 0.6 | 3.29 (± 0.12) | **2.86 (± 0.10)** | 1.54 (± 0.64) | **1.48 (± 0.70)** | 13.24 (± 2.03) | 8.22 (± 1.31) |
| | | 0.9 | 3.29 (± 0.11) | **2.09 (± 0.08)** | 1.69 (± 0.69) | **1.60 (± 0.74)** | 13.12 (± 2.34) | 12.64 (± 0.98) |
| $\mathbf{A}_3(t)$ | 1000 | 0.0 | **6.49 (± 0.22)** | 6.60 (± 0.23) | **2.21 (± 0.76)** | 2.27 (± 0.76) | 17.16 (± 2.64) | 2.85 (± 2.21) |
| | | 0.3 | **6.52 (± 0.18)** | 6.56 (± 0.17) | **2.46 (± 1.11)** | 2.53 (± 1.09) | 17.08 (± 3.21) | 11.00 (± 3.08) |
| | | 0.6 | 6.53 (± 0.19) | **6.29 (± 0.16)** | **2.70 (± 1.18)** | 2.73 (± 1.16) | 17.15 (± 3.26) | 22.32 (± 2.77) |
| | | 0.9 | 6.54 (± 0.22) | **5.75 (± 0.15)** | 2.95 (± 1.21) | **2.89 (± 1.22)** | 17.24 (± 3.35) | 33.66 (± 2.46) |
| | 2000 | 0.0 | **5.04 (± 0.11)** | 5.14 (± 0.12) | 1.49 (± 0.56) | **1.48 (± 0.58)** | 9.82 (± 1.17) | 1.33 (± 0.98) |
| | | 0.3 | **5.05 (± 0.12)** | 5.09 (± 0.12) | 1.41 (± 0.59) | **1.38 (± 0.61)** | 10.29 (± 1.30) | 8.32 (± 1.90) |
| | | 0.6 | 5.06 (± 0.13) | **4.89 (± 0.11)** | 1.55 (± 0.65) | **1.49 (± 0.66)** | 10.41 (± 1.36) | 16.81 (± 1.83) |
| | | 0.9 | 5.06 (± 0.13) | **4.50 (± 0.09)** | 1.67 (± 0.70) | **1.58 (± 0.70)** | 10.57 (± 1.48) | 25.32 (± 1.67) |
| $\mathbf{A}_4(t)$ | 1000 | 0.0 | **4.87 (± 0.19)** | 4.88 (± 0.20) | **2.09 (± 0.76)** | 2.13 (± 0.75) | 31.77 (± 2.69) | 2.85 (± 1.97) |
| | | 0.3 | 4.85 (± 0.18) | **4.73 (± 0.16)** | **2.43 (± 1.01)** | 2.47 (± 1.07) | 31.68 (± 3.47) | 6.83 (± 2.97) |
| | | 0.6 | 4.85 (± 0.21) | **4.28 (± 0.15)** | **2.65 (± 1.10)** | 2.67 (± 1.14) | 31.64 (± 3.69) | 13.98 (± 2.37) |
| | | 0.9 | 4.86 (± 0.24) | **3.20 (± 0.16)** | 2.88 (± 1.13) | **2.82 (± 1.15)** | 31.59 (± 3.97) | 21.14 (± 1.64) |
| | 2000 | 0.0 | **3.02 (± 0.09)** | 3.02 (± 0.09) | 1.26 (± 0.45) | **1.26 (± 0.45)** | 15.12 (± 1.87) | 1.44 (± 0.78) |
| | | 0.3 | 3.03 (± 0.09) | **2.91 (± 0.09)** | 1.36 (± 0.58) | **1.33 (± 0.60)** | 15.04 (± 1.85) | 3.57 (± 1.45) |
| | | 0.6 | 3.04 (± 0.09) | **2.54 (± 0.08)** | 1.51 (± 0.64) | **1.45 (± 0.66)** | 14.95 (± 2.06) | 7.33 (± 1.30) |
| | | 0.9 | 3.04 (± 0.09) | **1.63 (± 0.06)** | 1.65 (± 0.69) | **1.54 (± 0.71)** | 14.85 (± 2.35) | 11.21 (± 0.88) |
| $\mathbf{A}_5(t)$ | 1000 | 0.0 | **7.88 (± 0.22)** | 8.19 (± 0.24) | 2.56 (± 1.05) | **2.49 (± 1.00)** | 28.32 (± 2.37) | 3.50 (± 2.04) |
| | | 0.3 | **7.84 (± 0.26)** | 8.09 (± 0.26) | **2.63 (± 0.83)** | 2.71 (± 0.96) | 28.28 (± 2.54) | 14.55 (± 3.56) |
| | | 0.6 | **7.85 (± 0.27)** | 7.89 (± 0.26) | **2.85 (± 0.88)** | 2.93 (± 1.03) | 28.21 (± 2.58) | 29.92 (± 3.58) |
| | | 0.9 | 7.85 (± 0.30) | **7.48 (± 0.28)** | **3.08 (± 0.93)** | 3.13 (± 1.11) | 28.19 (± 2.66) | 45.30 (± 3.57) |
| | 2000 | 0.0 | **6.46 (± 0.14)** | 6.77 (± 0.14) | 1.59 (± 0.55) | **1.53 (± 0.50)** | 38.31 (± 2.77) | 2.00 (± 1.43) |
| | | 0.3 | **6.48 (± 0.13)** | 6.73 (± 0.13) | 1.66 (± 0.63) | **1.61 (± 0.55)** | 37.98 (± 2.08) | 13.04 (± 2.74) |
| | | 0.6 | **6.49 (± 0.13)** | 6.57 (± 0.13) | 1.79 (± 0.67) | **1.74 (± 0.59)** | 38.02 (± 2.04) | 26.38 (± 2.59) |
| | | 0.9 | 6.48 (± 0.13) | **6.24 (± 0.13)** | 1.93 (± 0.71) | **1.85 (± 0.61)** | 38.11 (± 2.09) | 39.79 (± 2.32) |
| $\mathbf{A}_6(t)$ | 1000 | 0.0 | **6.18 (± 0.21)** | 6.26 (± 0.21) | **2.10 (± 0.84)** | 2.15 (± 0.81) | 20.74 (± 2.87) | 2.91 (± 2.08) |
| | | 0.3 | 6.13 (± 0.21) | **6.12 (± 0.20)** | **2.62 (± 1.01)** | 2.64 (± 1.05) | 20.63 (± 3.74) | 9.38 (± 3.13) |
| | | 0.6 | 6.12 (± 0.23) | **5.81 (± 0.20)** | 2.83 (± 1.12) | **2.82 (± 1.13)** | 20.62 (± 3.94) | 19.66 (± 2.69) |
| | | 0.9 | 6.11 (± 0.25) | **5.18 (± 0.20)** | 3.04 (± 1.19) | **2.96 (± 1.18)** | 20.62 (± 4.18) | 29.96 (± 2.07) |
| | 2000 | 0.0 | **4.59 (± 0.10)** | 4.65 (± 0.10) | 1.17 (± 0.45) | **1.17 (± 0.45)** | 6.90 (± 1.09) | 1.67 (± 1.00) |
| | | 0.3 | **4.61 (± 0.13)** | 4.61 (± 0.12) | 1.55 (± 0.64) | **1.54 (± 0.67)** | 7.11 (± 1.22) | 6.86 (± 1.85) |
| | | 0.6 | 4.61 (± 0.13) | **4.40 (± 0.11)** | 1.71 (± 0.69) | **1.66 (± 0.74)** | 7.16 (± 1.31) | 14.36 (± 1.69) |
| | | 0.9 | 4.60 (± 0.14) | **3.95 (± 0.08)** | 1.86 (± 0.73) | **1.79 (± 0.78)** | 7.26 (± 1.44) | 21.94 (± 1.52) |

Table 2: MAE errors ($\times 10^{-2}$) for Case 1(b) - $\sigma_1(t) = 1 + 0.3\sin(2\pi t)$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 1000 | 0.0 | **5.05 (± 0.18)** | 5.05 (± 0.18) | 2.12 (± 0.80) | **2.05 (± 0.75)** | 30.43 (± 2.75) | 2.93 (± 1.94) |
| | | 0.3 | 5.03 (± 0.20) | **4.92 (± 0.18)** | **2.41 (± 1.03)** | 2.45 (± 1.04) | 30.27 (± 3.47) | 7.63 (± 3.19) |
| | | 0.6 | 5.03 (± 0.23) | **4.51 (± 0.17)** | **2.63 (± 1.10)** | 2.65 (± 1.10) | 30.26 (± 3.76) | 15.23 (± 2.67) |
| | | 0.9 | 5.04 (± 0.27) | **3.56 (± 0.18)** | 2.86 (± 1.14) | **2.79 (± 1.12)** | 30.23 (± 4.12) | 22.80 (± 2.02) |
| | 2000 | 0.0 | 3.29 (± 0.10) | **3.29 (± 0.10)** | 1.24 (± 0.46) | **1.21 (± 0.42)** | 13.47 (± 1.92) | 1.60 (± 0.90) |
| | | 0.3 | 3.31 (± 0.11) | **3.20 (± 0.11)** | 1.39 (± 0.61) | **1.34 (± 0.58)** | 13.41 (± 1.93) | 4.31 (± 1.55) |
| | | 0.6 | 3.32 (± 0.11) | **2.88 (± 0.10)** | 1.53 (± 0.67) | **1.44 (± 0.65)** | 13.46 (± 2.10) | 8.63 (± 1.31) |
| | | 0.9 | 3.32 (± 0.11) | **2.15 (± 0.07)** | 1.67 (± 0.72) | **1.53 (± 0.69)** | 13.57 (± 2.25) | 12.97 (± 0.98) |
| $\mathbf{A}_2(t)$ | 1000 | 0.0 | **5.05 (± 0.21)** | 5.07 (± 0.21) | 2.11 (± 0.82) | **2.06 (± 0.75)** | 30.43 (± 2.82) | 3.14 (± 2.03) |
| | | 0.3 | 4.99 (± 0.18) | **4.89 (± 0.17)** | **2.42 (± 1.01)** | 2.44 (± 1.02) | 30.32 (± 3.54) | 6.97 (± 3.07) |
| | | 0.6 | 4.99 (± 0.20) | **4.47 (± 0.16)** | 2.66 (± 1.09) | **2.63 (± 1.11)** | 30.32 (± 3.83) | 14.55 (± 2.42) |
| | | 0.9 | 4.99 (± 0.22) | **3.49 (± 0.15)** | 2.88 (± 1.15) | **2.78 (± 1.12)** | 30.30 (± 4.18) | 22.12 (± 1.51) |
| | 2000 | 0.0 | **3.26 (± 0.08)** | 3.27 (± 0.08) | 1.19 (± 0.46) | **1.18 (± 0.44)** | 13.55 (± 1.99) | 1.77 (± 1.12) |
| | | 0.3 | 3.26 (± 0.12) | **3.17 (± 0.12)** | 1.38 (± 0.62) | **1.35 (± 0.58)** | 13.41 (± 1.94) | 4.06 (± 1.53) |
| | | 0.6 | 3.26 (± 0.12) | **2.84 (± 0.10)** | 1.54 (± 0.67) | **1.46 (± 0.65)** | 13.41 (± 2.12) | 8.36 (± 1.28) |
| | | 0.9 | 3.26 (± 0.11) | **2.07 (± 0.08)** | 1.69 (± 0.71) | **1.56 (± 0.70)** | 13.45 (± 2.32) | 12.69 (± 1.00) |
| $\mathbf{A}_3(t)$ | 1000 | 0.0 | **6.48 (± 0.22)** | 6.59 (± 0.23) | 2.22 (± 0.79) | **2.21 (± 0.75)** | 19.24 (± 2.34) | 2.95 (± 2.18) |
| | | 0.3 | **6.51 (± 0.18)** | 6.54 (± 0.17) | **2.43 (± 1.08)** | 2.49 (± 1.08) | 19.03 (± 2.81) | 11.20 (± 3.26) |
| | | 0.6 | 6.52 (± 0.20) | **6.28 (± 0.17)** | **2.67 (± 1.15)** | 2.69 (± 1.15) | 19.03 (± 3.09) | 22.52 (± 2.99) |
| | | 0.9 | 6.53 (± 0.22) | **5.74 (± 0.16)** | 2.91 (± 1.19) | **2.86 (± 1.20)** | 19.06 (± 3.45) | 33.85 (± 2.68) |
| | 2000 | 0.0 | **5.04 (± 0.11)** | 5.14 (± 0.12) | 1.52 (± 0.55) | **1.47 (± 0.56)** | 10.30 (± 1.13) | 1.39 (± 1.04) |
| | | 0.3 | **5.05 (± 0.12)** | 5.08 (± 0.12) | 1.41 (± 0.58) | **1.35 (± 0.61)** | 10.84 (± 1.29) | 8.51 (± 1.89) |
| | | 0.6 | 5.06 (± 0.13) | **4.89 (± 0.11)** | 1.54 (± 0.63) | **1.46 (± 0.66)** | 10.99 (± 1.33) | 16.96 (± 1.81) |
| | | 0.9 | 5.06 (± 0.13) | **4.49 (± 0.09)** | 1.67 (± 0.68) | **1.56 (± 0.70)** | 11.22 (± 1.37) | 25.42 (± 1.66) |
| $\mathbf{A}_4(t)$ | 1000 | 0.0 | **4.85 (± 0.20)** | 4.86 (± 0.20) | 2.12 (± 0.78) | **2.05 (± 0.71)** | 31.88 (± 2.79) | 2.95 (± 2.01) |
| | | 0.3 | 4.82 (± 0.18) | **4.70 (± 0.16)** | **2.39 (± 1.03)** | 2.42 (± 1.05) | 31.72 (± 3.45) | 6.98 (± 3.13) |
| | | 0.6 | 4.82 (± 0.21) | **4.26 (± 0.15)** | 2.62 (± 1.10) | **2.61 (± 1.12)** | 31.71 (± 3.75) | 14.11 (± 2.52) |
| | | 0.9 | 4.82 (± 0.24) | **3.18 (± 0.15)** | 2.85 (± 1.14) | **2.77 (± 1.13)** | 31.69 (± 4.10) | 21.23 (± 1.76) |
| | 2000 | 0.0 | **3.00 (± 0.09)** | 3.00 (± 0.09) | 1.25 (± 0.47) | **1.23 (± 0.43)** | 15.24 (± 1.94) | 1.65 (± 0.92) |
| | | 0.3 | 3.01 (± 0.09) | **2.89 (± 0.09)** | 1.36 (± 0.59) | **1.31 (± 0.57)** | 15.15 (± 1.94) | 3.71 (± 1.47) |
| | | 0.6 | 3.02 (± 0.09) | **2.52 (± 0.08)** | 1.51 (± 0.65) | **1.42 (± 0.64)** | 15.14 (± 2.11) | 7.47 (± 1.21) |
| | | 0.9 | 3.02 (± 0.09) | **1.62 (± 0.06)** | 1.65 (± 0.70) | **1.51 (± 0.68)** | 15.15 (± 2.33) | 11.25 (± 0.85) |
| $\mathbf{A}_5(t)$ | 1000 | 0.0 | **7.85 (± 0.23)** | 8.15 (± 0.25) | 2.57 (± 1.07) | **2.44 (± 0.94)** | 26.70 (± 2.27) | 3.61 (± 2.24) |
| | | 0.3 | **7.81 (± 0.25)** | 8.05 (± 0.25) | 2.63 (± 0.82) | **2.60 (± 0.94)** | 26.69 (± 2.51) | 14.59 (± 3.55) |
| | | 0.6 | **7.81 (± 0.26)** | 7.85 (± 0.25) | 2.85 (± 0.86) | **2.82 (± 1.02)** | 26.63 (± 2.58) | 29.96 (± 3.56) |
| | | 0.9 | 7.81 (± 0.28) | **7.45 (± 0.27)** | 3.08 (± 0.90) | **3.02 (± 1.08)** | 26.58 (± 2.69) | 45.32 (± 3.51) |
| | 2000 | 0.0 | **6.44 (± 0.13)** | 6.74 (± 0.14) | 1.58 (± 0.53) | **1.48 (± 0.47)** | 38.26 (± 2.98) | 2.13 (± 1.50) |
| | | 0.3 | **6.46 (± 0.13)** | 6.70 (± 0.13) | 1.67 (± 0.63) | **1.57 (± 0.56)** | 37.88 (± 2.37) | 13.22 (± 2.83) |
| | | 0.6 | **6.46 (± 0.13)** | 6.54 (± 0.13) | 1.80 (± 0.67) | **1.69 (± 0.60)** | 37.88 (± 2.32) | 26.55 (± 2.68) |
| | | 0.9 | 6.46 (± 0.13) | **6.22 (± 0.13)** | 1.93 (± 0.71) | **1.80 (± 0.62)** | 37.90 (± 2.35) | 39.91 (± 2.40) |
| $\mathbf{A}_6(t)$ | 1000 | 0.0 | **6.16 (± 0.21)** | 6.23 (± 0.22) | 2.11 (± 0.87) | **2.11 (± 0.79)** | 20.88 (± 2.96) | 3.10 (± 2.08) |
| | | 0.3 | 6.10 (± 0.21) | **6.09 (± 0.20)** | 2.62 (± 1.05) | **2.57 (± 1.01)** | 20.76 (± 3.59) | 9.42 (± 3.30) |
| | | 0.6 | 6.09 (± 0.23) | **5.78 (± 0.20)** | 2.83 (± 1.15) | **2.75 (± 1.09)** | 20.79 (± 3.83) | 19.70 (± 2.82) |
| | | 0.9 | 6.08 (± 0.24) | **5.16 (± 0.20)** | 3.04 (± 1.23) | **2.88 (± 1.15)** | 20.80 (± 4.15) | 29.97 (± 2.13) |
| | 2000 | 0.0 | **4.58 (± 0.10)** | 4.64 (± 0.10) | 1.16 (± 0.45) | **1.16 (± 0.44)** | 6.98 (± 1.29) | 1.80 (± 1.19) |
| | | 0.3 | **4.58 (± 0.13)** | 4.59 (± 0.12) | 1.53 (± 0.69) | **1.52 (± 0.64)** | 7.08 (± 1.29) | 7.02 (± 1.88) |
| | | 0.6 | 4.59 (± 0.13) | **4.38 (± 0.11)** | 1.68 (± 0.75) | **1.64 (± 0.70)** | 7.21 (± 1.49) | 14.49 (± 1.74) |
| | | 0.9 | 4.58 (± 0.14) | **3.94 (± 0.08)** | 1.84 (± 0.79) | **1.75 (± 0.74)** | 7.38 (± 1.73) | 22.00 (± 1.61) |

Table 3: MAE errors ($\times 10^{-2}$) for Case 1(c) - $\sigma_2(t) = 1 + 0.25\cos(2\pi t) + 0.1\sin(4\pi t)$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 1000 | 0.0 | **4.28 (± 0.14)** | 4.29 (± 0.14) | 1.82 (± 0.68) | **1.78 (± 0.66)** | 20.98 (± 2.10) | 2.79 (± 1.96) |
| | | 0.3 | 4.28 (± 0.18) | **4.19 (± 0.16)** | **2.08 (± 0.85)** | 2.12 (± 0.89) | 20.91 (± 2.62) | 7.39 (± 3.17) |
| | | 0.6 | 4.28 (± 0.20) | **3.84 (± 0.15)** | **2.27 (± 0.92)** | 2.29 (± 0.94) | 20.89 (± 2.80) | 14.77 (± 2.63) |
| | | 0.9 | 4.28 (± 0.23) | **3.06 (± 0.15)** | 2.45 (± 0.96) | **2.40 (± 0.96)** | 20.85 (± 3.03) | 22.15 (± 1.96) |
| | 2000 | 0.0 | **2.85 (± 0.09)** | 2.86 (± 0.09) | 1.09 (± 0.38) | **1.04 (± 0.38)** | 8.56 (± 1.44) | 1.51 (± 0.91) |
| | | 0.3 | 2.87 (± 0.09) | **2.79 (± 0.09)** | 1.23 (± 0.52) | **1.13 (± 0.51)** | 8.47 (± 1.43) | 4.17 (± 1.73) |
| | | 0.6 | 2.88 (± 0.09) | **2.53 (± 0.09)** | 1.35 (± 0.57) | **1.23 (± 0.56)** | 8.41 (± 1.58) | 8.51 (± 1.42) |
| | | 0.9 | 2.88 (± 0.10) | **1.96 (± 0.06)** | 1.48 (± 0.61) | **1.31 (± 0.60)** | 8.37 (± 1.77) | 12.91 (± 1.01) |
| $\mathbf{A}_2(t)$ | 1000 | 0.0 | **4.29 (± 0.17)** | 4.31 (± 0.17) | 1.83 (± 0.69) | **1.77 (± 0.67)** | 21.00 (± 2.15) | 3.03 (± 2.04) |
| | | 0.3 | 4.24 (± 0.15) | **4.16 (± 0.14)** | **2.08 (± 0.83)** | 2.13 (± 0.87) | 20.97 (± 2.70) | 6.68 (± 2.98) |
| | | 0.6 | 4.24 (± 0.17) | **3.81 (± 0.13)** | **2.27 (± 0.89)** | 2.30 (± 0.93) | 20.95 (± 2.87) | 14.04 (± 2.36) |
| | | 0.9 | 4.23 (± 0.18) | **3.00 (± 0.13)** | 2.48 (± 0.92) | **2.41 (± 0.97)** | 20.91 (± 3.09) | 21.43 (± 1.49) |
| | 2000 | 0.0 | **2.83 (± 0.07)** | 2.84 (± 0.07) | 1.04 (± 0.40) | **1.01 (± 0.39)** | 8.68 (± 1.45) | 1.75 (± 1.15) |
| | | 0.3 | 2.83 (± 0.10) | **2.76 (± 0.10)** | 1.22 (± 0.53) | **1.15 (± 0.51)** | 8.56 (± 1.41) | 3.87 (± 1.72) |
| | | 0.6 | 2.83 (± 0.10) | **2.49 (± 0.08)** | 1.36 (± 0.56) | **1.25 (± 0.58)** | 8.49 (± 1.58) | 8.19 (± 1.38) |
| | | 0.9 | 2.83 (± 0.10) | **1.87 (± 0.06)** | 1.49 (± 0.59) | **1.34 (± 0.61)** | 8.42 (± 1.81) | 12.56 (± 0.99) |
| $\mathbf{A}_3(t)$ | 1000 | 0.0 | **5.84 (± 0.19)** | 5.96 (± 0.20) | **1.94 (± 0.68)** | 1.95 (± 0.68) | 9.43 (± 1.33) | 3.00 (± 2.14) |
| | | 0.3 | **5.88 (± 0.15)** | 5.94 (± 0.15) | **2.12 (± 0.98)** | 2.15 (± 0.90) | 9.57 (± 1.61) | 11.74 (± 3.09) |
| | | 0.6 | 5.88 (± 0.16) | **5.74 (± 0.14)** | 2.32 (± 1.05) | **2.32 (± 0.95)** | 9.61 (± 1.60) | 23.73 (± 2.84) |
| | | 0.9 | 5.89 (± 0.17) | **5.34 (± 0.13)** | 2.54 (± 1.06) | **2.45 (± 1.00)** | 9.69 (± 1.68) | 35.74 (± 2.62) |
| | 2000 | 0.0 | **4.72 (± 0.10)** | 4.84 (± 0.11) | 1.36 (± 0.51) | **1.29 (± 0.51)** | 10.46 (± 0.95) | 1.38 (± 1.05) |
| | | 0.3 | **4.72 (± 0.11)** | 4.80 (± 0.10) | 1.28 (± 0.50) | **1.17 (± 0.52)** | 10.69 (± 1.10) | 9.40 (± 2.06) |
| | | 0.6 | 4.73 (± 0.12) | **4.65 (± 0.10)** | 1.39 (± 0.55) | **1.26 (± 0.56)** | 10.78 (± 1.15) | 18.86 (± 1.97) |
| | | 0.9 | 4.73 (± 0.12) | **4.37 (± 0.08)** | 1.50 (± 0.59) | **1.34 (± 0.60)** | 10.92 (± 1.25) | 28.35 (± 1.83) |
| $\mathbf{A}_4(t)$ | 1000 | 0.0 | **4.07 (± 0.15)** | 4.07 (± 0.16) | 1.81 (± 0.67) | **1.79 (± 0.64)** | 22.43 (± 2.13) | 2.89 (± 1.95) |
| | | 0.3 | 4.05 (± 0.15) | **3.95 (± 0.14)** | **2.07 (± 0.86)** | 2.09 (± 0.89) | 22.37 (± 2.61) | 6.57 (± 3.04) |
| | | 0.6 | 4.06 (± 0.18) | **3.57 (± 0.13)** | **2.26 (± 0.93)** | 2.27 (± 0.94) | 22.34 (± 2.78) | 13.34 (± 2.41) |
| | | 0.9 | 4.05 (± 0.20) | **2.65 (± 0.13)** | 2.46 (± 0.96) | **2.38 (± 0.96)** | 22.31 (± 3.01) | 20.13 (± 1.62) |
| | 2000 | 0.0 | 2.55 (± 0.08) | **2.55 (± 0.08)** | 1.10 (± 0.40) | **1.05 (± 0.38)** | 10.37 (± 1.44) | 1.56 (± 0.90) |
| | | 0.3 | 2.56 (± 0.08) | **2.45 (± 0.08)** | 1.19 (± 0.50) | **1.11 (± 0.50)** | 10.27 (± 1.46) | 3.45 (± 1.54) |
| | | 0.6 | 2.56 (± 0.08) | **2.15 (± 0.07)** | 1.33 (± 0.55) | **1.21 (± 0.55)** | 10.21 (± 1.61) | 7.01 (± 1.30) |
| | | 0.9 | 2.56 (± 0.08) | **1.39 (± 0.05)** | 1.45 (± 0.59) | **1.28 (± 0.59)** | 10.14 (± 1.82) | 10.67 (± 0.83) |
| $\mathbf{A}_5(t)$ | 1000 | 0.0 | **7.99 (± 0.23)** | 8.31 (± 0.25) | 2.56 (± 1.06) | **2.50 (± 1.00)** | 32.41 (± 2.33) | 3.50 (± 2.03) |
| | | 0.3 | **7.96 (± 0.26)** | 8.21 (± 0.26) | **2.64 (± 0.83)** | 2.69 (± 0.96) | 32.40 (± 2.47) | 14.82 (± 3.62) |
| | | 0.6 | **7.96 (± 0.28)** | 8.00 (± 0.27) | **2.87 (± 0.90)** | 2.92 (± 1.03) | 32.35 (± 2.51) | 30.51 (± 3.66) |
| | | 0.9 | 7.97 (± 0.30) | **7.60 (± 0.29)** | **3.09 (± 0.96)** | 3.12 (± 1.09) | 32.35 (± 2.59) | 46.21 (± 3.67) |
| | 2000 | 0.0 | **6.52 (± 0.13)** | 6.83 (± 0.14) | 1.58 (± 0.56) | **1.53 (± 0.49)** | 40.76 (± 2.52) | 1.90 (± 1.35) |
| | | 0.3 | **6.55 (± 0.13)** | 6.80 (± 0.13) | 1.67 (± 0.62) | **1.61 (± 0.58)** | 40.52 (± 1.85) | 13.24 (± 2.74) |
| | | 0.6 | **6.55 (± 0.13)** | 6.63 (± 0.13) | 1.80 (± 0.65) | **1.74 (± 0.62)** | 40.56 (± 1.81) | 26.77 (± 2.60) |
| | | 0.9 | 6.55 (± 0.14) | **6.29 (± 0.13)** | 1.93 (± 0.69) | **1.85 (± 0.64)** | 40.62 (± 1.89) | 40.36 (± 2.34) |
| $\mathbf{A}_6(t)$ | 1000 | 0.0 | **6.26 (± 0.22)** | 6.34 (± 0.22) | **2.08 (± 0.86)** | 2.14 (± 0.78) | 22.13 (± 2.88) | 3.02 (± 2.21) |
| | | 0.3 | 6.20 (± 0.21) | **6.19 (± 0.20)** | 2.64 (± 1.05) | **2.55 (± 1.01)** | 21.97 (± 3.67) | 9.59 (± 3.29) |
| | | 0.6 | 6.19 (± 0.23) | **5.88 (± 0.20)** | 2.86 (± 1.15) | **2.73 (± 1.10)** | 21.94 (± 3.87) | 20.22 (± 2.83) |
| | | 0.9 | 6.18 (± 0.26) | **5.26 (± 0.21)** | 3.07 (± 1.24) | **2.88 (± 1.14)** | 21.92 (± 4.10) | 30.87 (± 2.19) |
| | 2000 | 0.0 | **4.61 (± 0.10)** | 4.68 (± 0.10) | 1.18 (± 0.46) | **1.15 (± 0.44)** | 7.91 (± 1.13) | 1.58 (± 1.01) |
| | | 0.3 | **4.64 (± 0.14)** | 4.64 (± 0.12) | 1.54 (± 0.64) | **1.50 (± 0.68)** | 8.20 (± 1.19) | 7.03 (± 1.92) |
| | | 0.6 | 4.64 (± 0.14) | **4.42 (± 0.11)** | 1.69 (± 0.69) | **1.62 (± 0.74)** | 8.24 (± 1.26) | 14.72 (± 1.77) |
| | | 0.9 | 4.63 (± 0.14) | **3.97 (± 0.09)** | 1.84 (± 0.73) | **1.74 (± 0.78)** | 8.31 (± 1.40) | 22.50 (± 1.60) |

Table 4: MAE errors ($\times 10^{-2}$) for Case 1(d) - $\sigma_3(t) = 0.5 + 0.5\sin^2(\pi t)$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 1000 | 0.0 | **5.17 (± 0.18)** | 5.18 (± 0.19) | **2.08 (± 0.79)** | 2.09 (± 0.75) | 31.65 (± 2.68) | 2.92 (± 2.04) |
| | | 0.3 | 5.15 (± 0.21) | **5.03 (± 0.18)** | 2.45 (± 1.03) | **2.44 (± 1.05)** | 31.52 (± 3.55) | 7.70 (± 3.10) |
| | | 0.6 | 5.16 (± 0.24) | **4.62 (± 0.17)** | 2.68 (± 1.10) | **2.64 (± 1.11)** | 31.47 (± 3.79) | 15.65 (± 2.60) |
| | | 0.9 | 5.16 (± 0.28) | **3.67 (± 0.19)** | 2.90 (± 1.16) | **2.79 (± 1.12)** | 31.40 (± 4.05) | 23.60 (± 2.02) |
| | 2000 | 0.0 | 3.35 (± 0.10) | **3.35 (± 0.10)** | 1.25 (± 0.43) | **1.23 (± 0.43)** | 14.39 (± 1.76) | 1.31 (± 0.81) |
| | | 0.3 | 3.38 (± 0.11) | **3.27 (± 0.11)** | 1.38 (± 0.59) | **1.35 (± 0.60)** | 14.46 (± 1.71) | 4.27 (± 1.69) |
| | | 0.6 | 3.39 (± 0.11) | **2.94 (± 0.10)** | 1.53 (± 0.64) | **1.46 (± 0.66)** | 14.38 (± 1.97) | 8.81 (± 1.48) |
| | | 0.9 | 3.39 (± 0.11) | **2.19 (± 0.07)** | 1.67 (± 0.69) | **1.55 (± 0.70)** | 14.32 (± 2.29) | 13.44 (± 1.10) |
| $\mathbf{A}_2(t)$ | 1000 | 0.0 | **5.19 (± 0.22)** | 5.20 (± 0.22) | **2.08 (± 0.80)** | 2.11 (± 0.76) | 31.67 (± 2.75) | 3.07 (± 2.20) |
| | | 0.3 | 5.13 (± 0.18) | **5.02 (± 0.17)** | 2.46 (± 1.00) | **2.45 (± 1.01)** | 31.56 (± 3.61) | 7.11 (± 3.01) |
| | | 0.6 | 5.13 (± 0.20) | **4.60 (± 0.16)** | 2.69 (± 1.10) | **2.65 (± 1.10)** | 31.52 (± 3.83) | 15.04 (± 2.38) |
| | | 0.9 | 5.13 (± 0.23) | **3.62 (± 0.16)** | 2.92 (± 1.16) | **2.80 (± 1.11)** | 31.47 (± 4.09) | 23.00 (± 1.58) |
| | 2000 | 0.0 | 3.34 (± 0.08) | **3.33 (± 0.08)** | 1.20 (± 0.46) | **1.19 (± 0.45)** | 14.82 (± 1.71) | 1.45 (± 0.96) |
| | | 0.3 | 3.36 (± 0.12) | **3.25 (± 0.12)** | 1.39 (± 0.59) | **1.35 (± 0.63)** | 14.94 (± 1.66) | 4.05 (± 1.61) |
| | | 0.6 | 3.36 (± 0.12) | **2.91 (± 0.11)** | 1.54 (± 0.64) | **1.47 (± 0.70)** | 14.87 (± 1.93) | 8.58 (± 1.36) |
| | | 0.9 | 3.36 (± 0.12) | **2.12 (± 0.08)** | 1.69 (± 0.69) | **1.58 (± 0.74)** | 14.79 (± 2.25) | 13.19 (± 1.04) |
| $\mathbf{A}_3(t)$ | 1000 | 0.0 | **6.60 (± 0.23)** | 6.71 (± 0.24) | **2.21 (± 0.77)** | 2.23 (± 0.74) | 19.12 (± 2.45) | 2.91 (± 2.24) |
| | | 0.3 | **6.62 (± 0.19)** | 6.66 (± 0.17) | **2.48 (± 1.09)** | 2.49 (± 1.09) | 19.06 (± 3.18) | 11.24 (± 3.14) |
| | | 0.6 | 6.63 (± 0.20) | **6.39 (± 0.16)** | 2.72 (± 1.16) | **2.69 (± 1.17)** | 19.13 (± 3.23) | 22.90 (± 2.84) |
| | | 0.9 | 6.64 (± 0.22) | **5.86 (± 0.16)** | 2.96 (± 1.20) | **2.85 (± 1.23)** | 19.23 (± 3.26) | 34.59 (± 2.55) |
| | 2000 | 0.0 | **5.10 (± 0.11)** | 5.19 (± 0.12) | 1.48 (± 0.55) | **1.47 (± 0.57)** | 12.79 (± 1.33) | 1.30 (± 0.97) |
| | | 0.3 | **5.11 (± 0.12)** | 5.14 (± 0.12) | 1.40 (± 0.60) | **1.37 (± 0.60)** | 13.23 (± 1.41) | 8.47 (± 1.87) |
| | | 0.6 | 5.12 (± 0.13) | **4.94 (± 0.11)** | 1.53 (± 0.66) | **1.48 (± 0.64)** | 13.33 (± 1.53) | 17.14 (± 1.83) |
| | | 0.9 | 5.12 (± 0.13) | **4.53 (± 0.09)** | 1.65 (± 0.72) | **1.58 (± 0.68)** | 13.45 (± 1.70) | 25.86 (± 1.70) |
| $\mathbf{A}_4(t)$ | 1000 | 0.0 | **4.97 (± 0.21)** | 4.98 (± 0.21) | **2.08 (± 0.77)** | 2.10 (± 0.72) | 33.11 (± 2.70) | 2.93 (± 2.14) |
| | | 0.3 | 4.94 (± 0.19) | **4.82 (± 0.16)** | 2.43 (± 1.03) | **2.42 (± 1.04)** | 32.96 (± 3.54) | 7.05 (± 3.01) |
| | | 0.6 | 4.95 (± 0.21) | **4.37 (± 0.15)** | 2.67 (± 1.11) | **2.61 (± 1.13)** | 32.92 (± 3.77) | 14.53 (± 2.42) |
| | | 0.9 | 4.95 (± 0.25) | **3.30 (± 0.16)** | 2.90 (± 1.16) | **2.77 (± 1.13)** | 32.86 (± 4.03) | 22.04 (± 1.74) |
| | 2000 | 0.0 | **3.06 (± 0.09)** | 3.06 (± 0.09) | 1.26 (± 0.44) | **1.24 (± 0.45)** | 16.05 (± 1.84) | 1.38 (± 0.82) |
| | | 0.3 | 3.08 (± 0.10) | **2.96 (± 0.10)** | 1.35 (± 0.58) | **1.32 (± 0.59)** | 16.08 (± 1.78) | 3.71 (± 1.48) |
| | | 0.6 | 3.08 (± 0.09) | **2.59 (± 0.08)** | 1.50 (± 0.64) | **1.43 (± 0.65)** | 15.99 (± 2.05) | 7.66 (± 1.36) |
| | | 0.9 | 3.09 (± 0.09) | **1.66 (± 0.06)** | 1.64 (± 0.69) | **1.53 (± 0.69)** | 15.89 (± 2.39) | 11.73 (± 0.95) |
| $\mathbf{A}_5(t)$ | 1000 | 0.0 | **7.26 (± 0.19)** | 7.59 (± 0.22) | 2.34 (± 0.93) | **2.14 (± 0.85)** | 25.26 (± 2.29) | 3.63 (± 2.11) |
| | | 0.3 | **7.23 (± 0.23)** | 7.52 (± 0.24) | **2.31 (± 0.75)** | 2.33 (± 0.83) | 25.11 (± 2.60) | 15.73 (± 3.80) |
| | | 0.6 | **7.23 (± 0.25)** | 7.36 (± 0.25) | **2.51 (± 0.78)** | 2.51 (± 0.88) | 25.06 (± 2.68) | 32.31 (± 3.85) |
| | | 0.9 | 7.23 (± 0.26) | **7.08 (± 0.26)** | 2.69 (± 0.80) | **2.65 (± 0.96)** | 25.03 (± 2.79) | 48.89 (± 3.85) |
| | 2000 | 0.0 | **6.14 (± 0.13)** | 6.49 (± 0.14) | 1.47 (± 0.49) | **1.33 (± 0.44)** | 39.37 (± 2.68) | 2.28 (± 1.74) |
| | | 0.3 | **6.16 (± 0.12)** | 6.46 (± 0.12) | 1.52 (± 0.58) | **1.39 (± 0.45)** | 38.85 (± 2.12) | 14.64 (± 2.96) |
| | | 0.6 | **6.16 (± 0.12)** | 6.34 (± 0.12) | 1.63 (± 0.62) | **1.49 (± 0.48)** | 38.90 (± 2.11) | 29.59 (± 2.81) |
| | | 0.9 | 6.15 (± 0.13) | **6.12 (± 0.13)** | 1.74 (± 0.66) | **1.58 (± 0.49)** | 39.00 (± 2.17) | 44.58 (± 2.55) |
| $\mathbf{A}_6(t)$ | 1000 | 0.0 | **5.57 (± 0.18)** | 5.63 (± 0.18) | 1.88 (± 0.72) | **1.85 (± 0.69)** | 12.29 (± 1.89) | 2.83 (± 2.15) |
| | | 0.3 | **5.52 (± 0.18)** | 5.52 (± 0.18) | **2.27 (± 0.86)** | 2.32 (± 0.90) | 12.39 (± 2.44) | 9.89 (± 3.12) |
| | | 0.6 | 5.51 (± 0.19) | **5.28 (± 0.18)** | **2.45 (± 0.94)** | 2.47 (± 0.98) | 12.43 (± 2.53) | 20.64 (± 2.74) |
| | | 0.9 | 5.49 (± 0.21) | **4.80 (± 0.18)** | 2.63 (± 0.98) | **2.58 (± 1.03)** | 12.48 (± 2.66) | 31.40 (± 2.19) |
| | 2000 | 0.0 | **4.29 (± 0.09)** | 4.35 (± 0.10) | 1.04 (± 0.40) | **1.03 (± 0.38)** | 9.57 (± 1.10) | 1.83 (± 1.18) |
| | | 0.3 | **4.30 (± 0.11)** | 4.32 (± 0.10) | 1.39 (± 0.58) | **1.36 (± 0.56)** | 9.76 (± 1.20) | 7.66 (± 1.92) |
| | | 0.6 | 4.30 (± 0.11) | **4.16 (± 0.09)** | 1.52 (± 0.63) | **1.47 (± 0.61)** | 9.77 (± 1.36) | 15.93 (± 1.74) |
| | | 0.9 | 4.29 (± 0.11) | **3.83 (± 0.07)** | 1.65 (± 0.67) | **1.58 (± 0.65)** | 9.79 (± 1.54) | 24.27 (± 1.57) |

Below we present three examples, with fitted-versus-true curves for $\mathbf{A}_2$, $\mathbf{A}_4$, and $\mathbf{A}_6$, respectively. These plots show that our method is effective across a range of different $\mathbf{A}(t)$ configurations.

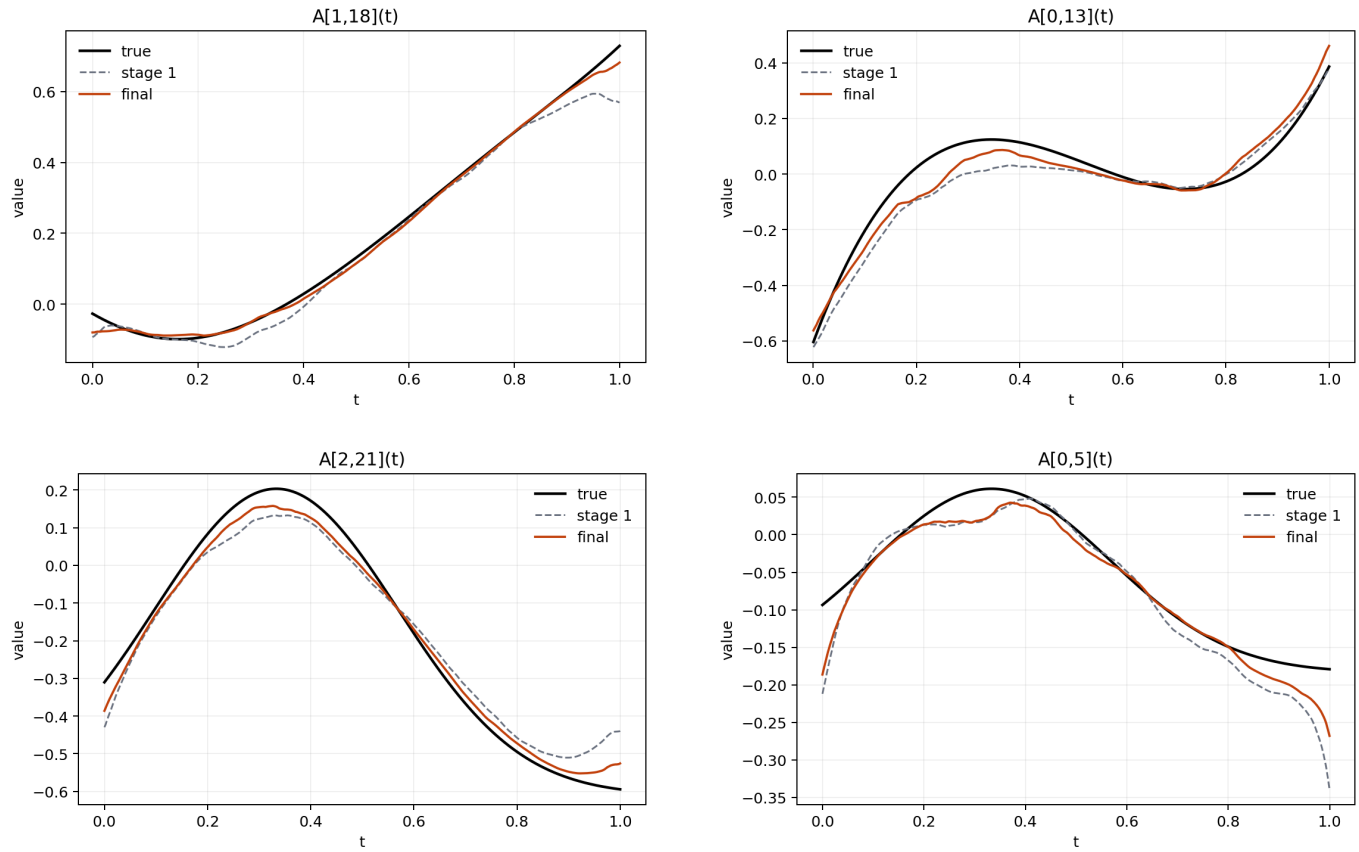
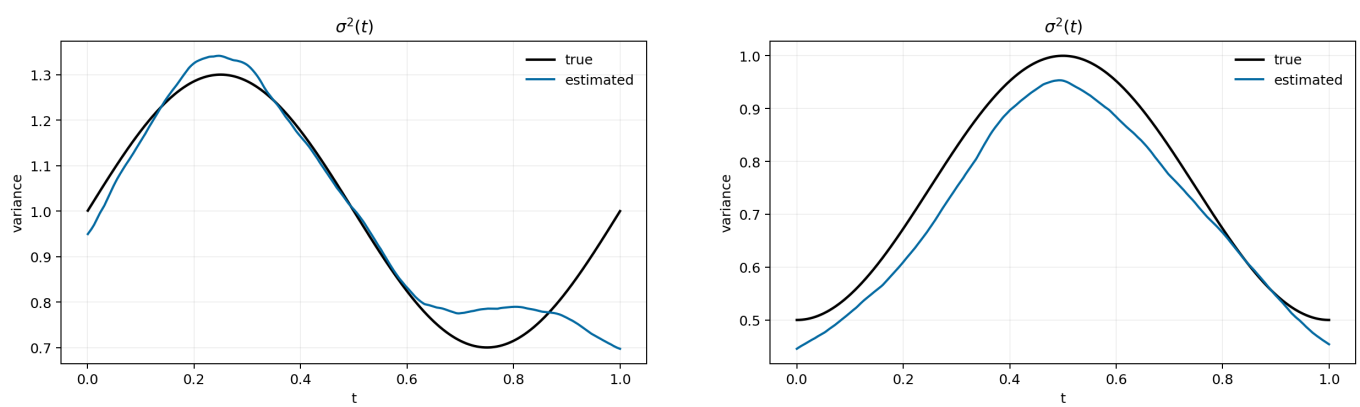



Figure 1: Selected estimated coefficient-function curves from Case 1.

We also present 3 examples of fitted-versus-true curves for $\sigma^2(t)$. They show that our method is effective for estimating the variance function across different settings.




(a) True $\sigma^2(t) = 1 + 0.3\sin(2\pi t)$.

(b) True $\sigma^2(t) = 0.5 + 0.5\sin^2(\pi t)$.

Figure 2: Selected estimated variance-function curves from Case 1.

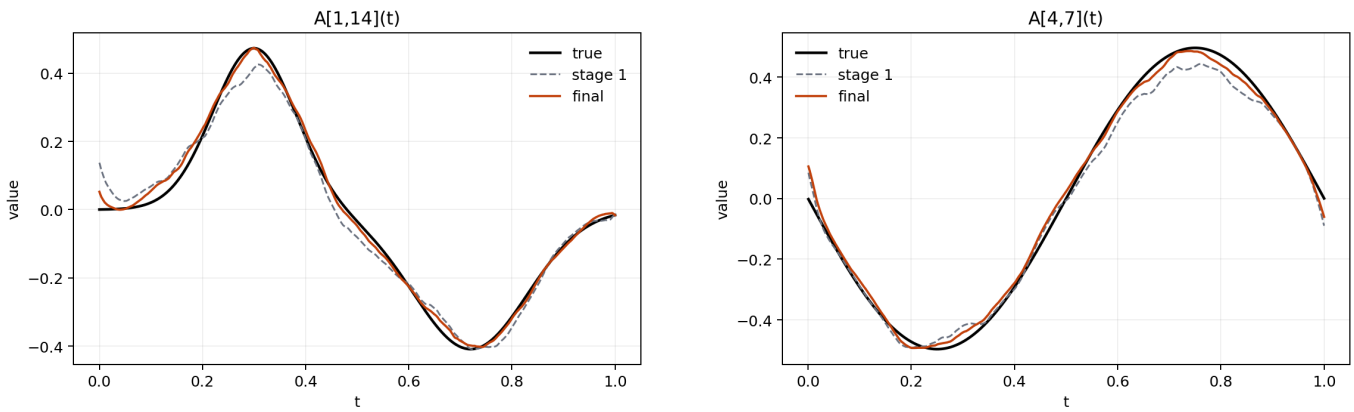
## Case 2: 3D Tensor Covariate

**DGP.** The tensor size is set to be $D = 3$, $p_1 = p_2 = p_3 = 48$, and split by $p'_1 = p'_2 = p'_3 = 16$ such that by $S^{(1)} = S^{(2)} = S^{(3)} = 3$. Set the rank $R = 6$, and the $s$-th block tensor for $j$-th eye of $i$-th patient is generated by:

$$\mathcal{X}_{ij}^{(s)} = \sum_{r=1}^{R} x^*_{ijrs} \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}, \quad \text{where} \quad x^*_{ijrs} \overset{\text{iid}}{\sim} \mathrm{N}(0,1).$$

and any $\mathbf{x}_r^{(sd)}$ and $\mathbf{x}_{r'}^{(sd)}$ are orthonormal and obtained from the QR decomposition of a random matrix. The coefficient function is set to be $\mathcal{A}^{(s)}(t) = \sum_{r=1}^{R} \alpha_{rs}(t) \mathbf{x}_r^{(s1)} \circ \mathbf{x}_r^{(s2)} \circ \mathbf{x}_r^{(s3)}$ where $\alpha_{rs}(t)$ has the same 4 choices as in Case 1. And everything else is the same.

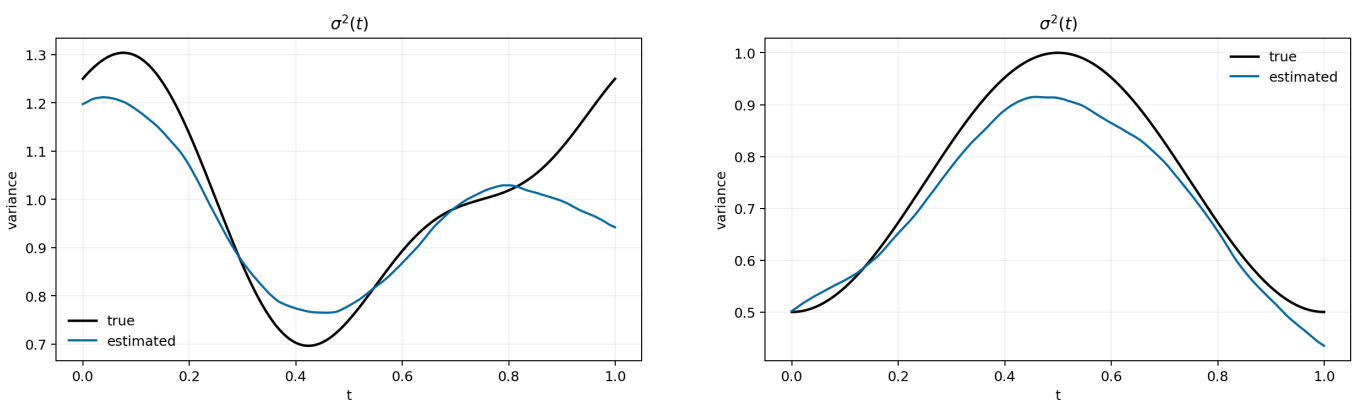
**Simulation Results.** We vary $n = 2000/5000$, coefficients $\alpha_{1rs}(t), \ldots, \alpha_{6rs}(t)$, and correlation $\rho = 0/0.3/0.6/0.9$. Tables 5–8 summarize the Case 2 results under the four variance settings. Figure 3 presents representative fitted-versus-true coefficient-function curves under $\mathbf{A}_5(t)$ and $\mathbf{A}_3(t)$, and Figure 4 presents two representative fitted-versus-true variance-function curves.




(a) Coefficient function under $\mathbf{A}_5(t)$.

(b) Coefficient function under $\mathbf{A}_3(t)$.

Figure 3: Representative fitted-versus-true coefficient-function curves from Case 2.




(a) $\sigma^2(t) = 1 + 0.25\cos(2\pi t) + 0.1\sin(4\pi t)$.

(b) $\sigma^2(t) = 0.5 + 0.5\sin^2(\pi t)$.

Figure 4: Representative fitted-versus-true variance-function curves from Case 2.

Table 5: MAE errors ($\times 10^{-2}$) for Case 2(a) - $\sigma_0^2(t) \equiv 1$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 2000 | 0.0 | **3.51 (± 0.10)** | 3.53 (± 0.10) | 1.55 (± 0.65) | **1.53 (± 0.66)** | 18.86 (± 2.34) | 1.96 (± 1.45) |
| | | 0.3 | 3.51 (± 0.10) | **3.44 (± 0.10)** | 1.70 (± 0.71) | **1.69 (± 0.71)** | 18.64 (± 2.18) | 6.54 (± 2.61) |
| | | 0.6 | 3.51 (± 0.10) | **3.17 (± 0.09)** | 1.86 (± 0.77) | **1.83 (± 0.75)** | 18.61 (± 2.62) | 13.45 (± 2.06) |
| | | 0.9 | 3.52 (± 0.11) | **2.58 (± 0.08)** | 2.02 (± 0.83) | **1.96 (± 0.77)** | 18.66 (± 2.97) | 20.39 (± 1.34) |
| | 5000 | 0.0 | **2.42 (± 0.06)** | 2.43 (± 0.06) | **0.90 (± 0.36)** | 0.91 (± 0.37) | 4.58 (± 0.92) | 1.15 (± 0.78) |
| | | 0.3 | 2.39 (± 0.06) | **2.35 (± 0.06)** | 0.95 (± 0.30) | **0.94 (± 0.30)** | 4.61 (± 0.98) | 3.99 (± 1.29) |
| | | 0.6 | 2.39 (± 0.06) | **2.20 (± 0.06)** | 1.04 (± 0.34) | **1.03 (± 0.32)** | 4.78 (± 1.16) | 7.73 (± 1.04) |
| | | 0.9 | 2.39 (± 0.06) | **1.89 (± 0.04)** | 1.14 (± 0.37) | **1.11 (± 0.34)** | 5.03 (± 1.35) | 11.42 (± 0.64) |
| $\mathbf{A}_2(t)$ | 2000 | 0.0 | **3.46 (± 0.10)** | 3.48 (± 0.10) | 1.52 (± 0.72) | **1.50 (± 0.73)** | 19.09 (± 2.34) | 1.68 (± 1.37) |
| | | 0.3 | 3.45 (± 0.09) | **3.39 (± 0.10)** | 1.75 (± 0.68) | **1.73 (± 0.67)** | 18.90 (± 2.18) | 6.28 (± 2.39) |
| | | 0.6 | 3.45 (± 0.09) | **3.11 (± 0.09)** | 1.91 (± 0.74) | **1.87 (± 0.74)** | 18.89 (± 2.55) | 13.04 (± 2.10) |
| | | 0.9 | 3.45 (± 0.10) | **2.47 (± 0.08)** | 2.06 (± 0.80) | **1.99 (± 0.78)** | 18.92 (± 2.91) | 19.93 (± 1.51) |
| | 5000 | 0.0 | **2.35 (± 0.05)** | 2.36 (± 0.05) | **0.91 (± 0.38)** | 0.91 (± 0.39) | 6.60 (± 0.93) | 1.21 (± 0.73) |
| | | 0.3 | 2.33 (± 0.05) | **2.29 (± 0.05)** | 0.92 (± 0.30) | **0.91 (± 0.29)** | 6.61 (± 0.96) | 3.91 (± 1.23) |
| | | 0.6 | 2.32 (± 0.05) | **2.12 (± 0.04)** | 1.02 (± 0.32) | **1.00 (± 0.31)** | 6.73 (± 1.09) | 7.63 (± 0.99) |
| | | 0.9 | 2.32 (± 0.05) | **1.74 (± 0.04)** | 1.12 (± 0.36) | **1.08 (± 0.33)** | 6.88 (± 1.27) | 11.32 (± 0.66) |
| $\mathbf{A}_3(t)$ | 2000 | 0.0 | **4.56 (± 0.11)** | 4.60 (± 0.11) | 1.55 (± 0.72) | **1.55 (± 0.71)** | 22.37 (± 2.16) | 1.89 (± 1.60) |
| | | 0.3 | 4.56 (± 0.11) | **4.52 (± 0.10)** | 1.76 (± 0.79) | **1.75 (± 0.82)** | 22.36 (± 2.18) | 9.36 (± 2.40) |
| | | 0.6 | 4.62 (± 0.12) | **4.25 (± 0.10)** | 1.93 (± 0.85) | **1.89 (± 0.91)** | 31.24 (± 2.40) | 18.48 (± 1.71) |
| | | 0.9 | 4.62 (± 0.13) | **3.63 (± 0.10)** | 2.06 (± 0.90) | **1.99 (± 0.94)** | 31.17 (± 2.80) | 28.21 (± 1.02) |
| | 5000 | 0.0 | **2.70 (± 0.05)** | 2.70 (± 0.05) | 0.93 (± 0.31) | **0.93 (± 0.33)** | 13.81 (± 1.23) | 1.10 (± 0.68) |
| | | 0.3 | 2.68 (± 0.06) | **2.61 (± 0.05)** | 0.95 (± 0.35) | **0.94 (± 0.35)** | 13.72 (± 1.23) | 4.90 (± 1.17) |
| | | 0.6 | 2.68 (± 0.06) | **2.36 (± 0.05)** | 1.06 (± 0.37) | **1.02 (± 0.37)** | 13.78 (± 1.33) | 9.50 (± 0.95) |
| | | 0.9 | 2.78 (± 0.06) | **1.68 (± 0.04)** | 1.16 (± 0.39) | **1.10 (± 0.36)** | 19.45 (± 1.49) | 14.72 (± 0.59) |
| $\mathbf{A}_4(t)$ | 2000 | 0.0 | **3.13 (± 0.09)** | 3.14 (± 0.09) | 1.51 (± 0.67) | **1.49 (± 0.67)** | 22.56 (± 2.39) | 1.83 (± 1.46) |
| | | 0.3 | 3.12 (± 0.08) | **3.03 (± 0.08)** | 1.68 (± 0.70) | **1.67 (± 0.71)** | 22.38 (± 2.22) | 5.44 (± 2.37) |
| | | 0.6 | 3.13 (± 0.09) | **2.71 (± 0.07)** | 1.85 (± 0.77) | **1.81 (± 0.77)** | 22.33 (± 2.70) | 11.14 (± 1.88) |
| | | 0.9 | 3.13 (± 0.10) | **1.91 (± 0.06)** | 2.00 (± 0.82) | **1.94 (± 0.80)** | 22.29 (± 3.24) | 16.95 (± 1.02) |
| | 5000 | 0.0 | **1.87 (± 0.04)** | 1.87 (± 0.04) | **0.91 (± 0.33)** | 0.91 (± 0.34) | 8.40 (± 1.33) | 1.13 (± 0.74) |
| | | 0.3 | 1.84 (± 0.05) | **1.78 (± 0.05)** | 0.93 (± 0.30) | **0.92 (± 0.30)** | 8.34 (± 1.30) | 2.81 (± 1.20) |
| | | 0.6 | 1.84 (± 0.06) | **1.57 (± 0.05)** | 1.03 (± 0.33) | **1.01 (± 0.32)** | 8.43 (± 1.43) | 5.30 (± 0.92) |
| | | 0.9 | 1.84 (± 0.06) | **1.06 (± 0.03)** | 1.13 (± 0.37) | **1.10 (± 0.34)** | 8.52 (± 1.65) | 7.72 (± 0.48) |
| $\mathbf{A}_5(t)$ | 2000 | 0.0 | **5.38 (± 0.13)** | 5.41 (± 0.13) | **1.69 (± 0.65)** | 1.69 (± 0.70) | 22.85 (± 1.37) | 2.04 (± 1.43) |
| | | 0.3 | 5.39 (± 0.15) | **5.36 (± 0.15)** | 1.95 (± 0.86) | **1.88 (± 0.93)** | 22.74 (± 1.56) | 11.82 (± 2.49) |
| | | 0.6 | 5.39 (± 0.16) | **5.12 (± 0.16)** | 2.09 (± 0.91) | **2.01 (± 1.01)** | 22.77 (± 1.78) | 24.36 (± 2.28) |
| | | 0.9 | 5.39 (± 0.18) | **4.66 (± 0.17)** | 2.21 (± 0.95) | **2.11 (± 1.05)** | 22.82 (± 2.03) | 36.95 (± 2.04) |
| | 5000 | 0.0 | **3.13 (± 0.05)** | 3.14 (± 0.05) | **0.94 (± 0.42)** | 0.95 (± 0.43) | 15.82 (± 1.17) | 1.26 (± 0.66) |
| | | 0.3 | 3.11 (± 0.05) | **3.04 (± 0.05)** | 1.00 (± 0.29) | **0.99 (± 0.30)** | 15.76 (± 1.25) | 6.00 (± 1.24) |
| | | 0.6 | 3.11 (± 0.06) | **2.77 (± 0.04)** | 1.09 (± 0.32) | **1.07 (± 0.31)** | 15.80 (± 1.34) | 11.97 (± 1.03) |
| | | 0.9 | 3.10 (± 0.06) | **2.16 (± 0.03)** | 1.19 (± 0.35) | **1.14 (± 0.33)** | 15.84 (± 1.49) | 17.89 (± 0.72) |
| $\mathbf{A}_6(t)$ | 2000 | 0.0 | **4.42 (± 0.10)** | 4.44 (± 0.11) | **1.60 (± 0.77)** | 1.62 (± 0.78) | 24.10 (± 2.05) | 1.86 (± 1.08) |
| | | 0.3 | 4.42 (± 0.11) | **4.36 (± 0.12)** | 1.76 (± 0.73) | **1.74 (± 0.74)** | 24.05 (± 2.19) | 8.50 (± 2.63) |
| | | 0.6 | 4.42 (± 0.11) | **4.10 (± 0.11)** | 1.92 (± 0.79) | **1.88 (± 0.80)** | 23.99 (± 2.60) | 17.91 (± 2.19) |
| | | 0.9 | 4.55 (± 0.13) | **3.51 (± 0.11)** | 2.05 (± 0.87) | **1.97 (± 0.88)** | 31.82 (± 2.90) | 27.33 (± 1.36) |
| | 5000 | 0.0 | **2.67 (± 0.05)** | 2.67 (± 0.05) | 0.93 (± 0.35) | **0.92 (± 0.35)** | 14.08 (± 1.23) | 1.17 (± 0.77) |
| | | 0.3 | 2.65 (± 0.05) | **2.58 (± 0.05)** | 0.97 (± 0.31) | **0.95 (± 0.32)** | 14.02 (± 1.25) | 4.64 (± 1.18) |
| | | 0.6 | 2.65 (± 0.05) | **2.33 (± 0.05)** | 1.07 (± 0.34) | **1.03 (± 0.33)** | 14.07 (± 1.36) | 9.18 (± 0.97) |
| | | 0.9 | 2.78 (± 0.06) | **1.67 (± 0.04)** | 1.17 (± 0.36) | **1.10 (± 0.35)** | 19.51 (± 1.51) | 14.59 (± 0.55) |

Table 6: MAE errors ($\times 10^{-2}$) for Case 2(b) - $\sigma_1^2(t) = 1 + 0.3\sin(2\pi t)$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 2000 | 0.0 | **3.52 (± 0.10)** | 3.52 (± 0.10) | 1.54 (± 0.66) | **1.52 (± 0.66)** | 22.39 (± 2.22) | 1.96 (± 1.43) |
| | | 0.3 | 3.51 (± 0.09) | **3.43 (± 0.09)** | 1.72 (± 0.69) | **1.63 (± 0.70)** | 22.33 (± 2.07) | 6.34 (± 2.58) |
| | | 0.6 | 3.51 (± 0.10) | **3.13 (± 0.09)** | 1.88 (± 0.74) | **1.76 (± 0.77)** | 22.27 (± 2.51) | 13.21 (± 1.97) |
| | | 0.9 | 3.52 (± 0.11) | **2.46 (± 0.08)** | 2.02 (± 0.80) | **1.89 (± 0.78)** | 22.23 (± 2.92) | 20.14 (± 1.26) |
| | 5000 | 0.0 | 2.28 (± 0.06) | **2.28 (± 0.06)** | 0.91 (± 0.36) | **0.90 (± 0.37)** | 7.11 (± 1.15) | 1.16 (± 0.78) |
| | | 0.3 | 2.24 (± 0.06) | **2.19 (± 0.06)** | 0.93 (± 0.30) | **0.92 (± 0.29)** | 7.10 (± 1.21) | 3.62 (± 1.23) |
| | | 0.6 | 2.24 (± 0.06) | **2.02 (± 0.05)** | 1.03 (± 0.33) | **1.01 (± 0.31)** | 7.20 (± 1.32) | 7.03 (± 0.99) |
| | | 0.9 | 2.24 (± 0.06) | **1.64 (± 0.04)** | 1.12 (± 0.36) | **1.10 (± 0.33)** | 7.36 (± 1.50) | 10.39 (± 0.61) |
| $\mathbf{A}_2(t)$ | 2000 | 0.0 | **3.46 (± 0.10)** | 3.48 (± 0.10) | 1.50 (± 0.70) | **1.47 (± 0.70)** | 22.57 (± 2.21) | 1.72 (± 1.38) |
| | | 0.3 | 3.46 (± 0.10) | **3.39 (± 0.10)** | 1.74 (± 0.66) | **1.64 (± 0.69)** | 22.53 (± 2.07) | 6.13 (± 2.33) |
| | | 0.6 | 3.46 (± 0.10) | **3.09 (± 0.09)** | 1.91 (± 0.71) | **1.78 (± 0.76)** | 22.45 (± 2.52) | 12.87 (± 2.01) |
| | | 0.9 | 3.46 (± 0.11) | **2.39 (± 0.08)** | 2.06 (± 0.77) | **1.90 (± 0.80)** | 22.39 (± 2.98) | 19.77 (± 1.44) |
| | 5000 | 0.0 | **2.22 (± 0.05)** | 2.23 (± 0.05) | 0.91 (± 0.36) | **0.90 (± 0.38)** | 6.84 (± 1.25) | 1.25 (± 0.73) |
| | | 0.3 | 2.20 (± 0.05) | **2.16 (± 0.04)** | 0.91 (± 0.31) | **0.90 (± 0.29)** | 6.79 (± 1.18) | 3.56 (± 1.18) |
| | | 0.6 | 2.19 (± 0.05) | **1.97 (± 0.04)** | 1.01 (± 0.34) | **0.99 (± 0.31)** | 6.90 (± 1.29) | 6.97 (± 0.94) |
| | | 0.9 | 2.19 (± 0.05) | **1.54 (± 0.03)** | 1.11 (± 0.36) | **1.07 (± 0.32)** | 7.05 (± 1.46) | 10.34 (± 0.62) |
| $\mathbf{A}_3(t)$ | 2000 | 0.0 | **4.56 (± 0.11)** | 4.58 (± 0.11) | 1.55 (± 0.70) | **1.50 (± 0.71)** | 22.24 (± 2.16) | 1.97 (± 1.52) |
| | | 0.3 | 4.56 (± 0.10) | **4.51 (± 0.10)** | 1.77 (± 0.77) | **1.72 (± 0.82)** | 22.39 (± 2.12) | 9.33 (± 2.34) |
| | | 0.6 | 4.61 (± 0.12) | **4.24 (± 0.10)** | 1.94 (± 0.83) | **1.85 (± 0.89)** | 31.19 (± 2.41) | 18.46 (± 1.65) |
| | | 0.9 | 4.62 (± 0.13) | **3.62 (± 0.09)** | 2.07 (± 0.87) | **1.94 (± 0.92)** | 31.08 (± 2.79) | 28.22 (± 1.05) |
| | 5000 | 0.0 | **2.69 (± 0.05)** | 2.69 (± 0.05) | 0.93 (± 0.31) | **0.92 (± 0.34)** | 13.72 (± 1.22) | 1.15 (± 0.76) |
| | | 0.3 | 2.68 (± 0.06) | **2.61 (± 0.05)** | 0.95 (± 0.35) | **0.93 (± 0.34)** | 13.66 (± 1.25) | 4.84 (± 1.14) |
| | | 0.6 | 2.68 (± 0.06) | **2.36 (± 0.05)** | 1.05 (± 0.38) | **1.01 (± 0.35)** | 13.71 (± 1.34) | 9.45 (± 0.94) |
| | | 0.9 | 2.77 (± 0.06) | **1.67 (± 0.04)** | 1.16 (± 0.39) | **1.08 (± 0.35)** | 19.35 (± 1.49) | 14.69 (± 0.59) |
| $\mathbf{A}_4(t)$ | 2000 | 0.0 | **3.25 (± 0.09)** | 3.26 (± 0.09) | 1.49 (± 0.68) | **1.48 (± 0.67)** | 24.79 (± 2.24) | 1.84 (± 1.49) |
| | | 0.3 | 3.25 (± 0.08) | **3.15 (± 0.08)** | 1.70 (± 0.69) | **1.61 (± 0.72)** | 24.76 (± 2.11) | 5.60 (± 2.34) |
| | | 0.6 | 3.25 (± 0.09) | **2.82 (± 0.07)** | 1.86 (± 0.75) | **1.75 (± 0.78)** | 24.68 (± 2.57) | 11.64 (± 1.82) |
| | | 0.9 | 3.26 (± 0.10) | **2.00 (± 0.06)** | 2.01 (± 0.80) | **1.88 (± 0.81)** | 24.61 (± 3.06) | 17.83 (± 1.03) |
| | 5000 | 0.0 | **1.89 (± 0.05)** | 1.89 (± 0.04) | 0.91 (± 0.33) | **0.90 (± 0.34)** | 9.48 (± 1.31) | 1.20 (± 0.72) |
| | | 0.3 | 1.86 (± 0.05) | **1.80 (± 0.05)** | 0.92 (± 0.31) | **0.91 (± 0.29)** | 9.44 (± 1.26) | 2.82 (± 1.16) |
| | | 0.6 | 1.86 (± 0.06) | **1.57 (± 0.05)** | 1.02 (± 0.34) | **1.00 (± 0.31)** | 9.53 (± 1.37) | 5.38 (± 0.90) |
| | | 0.9 | 1.86 (± 0.06) | **1.02 (± 0.03)** | 1.11 (± 0.37) | **1.08 (± 0.32)** | 9.64 (± 1.55) | 7.89 (± 0.50) |
| $\mathbf{A}_5(t)$ | 2000 | 0.0 | **5.34 (± 0.12)** | 5.38 (± 0.13) | 1.69 (± 0.63) | **1.64 (± 0.68)** | 20.41 (± 1.81) | 1.99 (± 1.48) |
| | | 0.3 | 5.36 (± 0.15) | **5.32 (± 0.15)** | 1.96 (± 0.82) | **1.81 (± 0.90)** | 20.48 (± 2.03) | 11.74 (± 2.52) |
| | | 0.6 | 5.36 (± 0.16) | **5.09 (± 0.16)** | 2.10 (± 0.87) | **1.94 (± 0.98)** | 20.44 (± 2.30) | 24.30 (± 2.31) |
| | | 0.9 | 5.36 (± 0.18) | **4.63 (± 0.16)** | 2.22 (± 0.91) | **2.03 (± 1.03)** | 20.40 (± 2.57) | 36.92 (± 2.09) |
| | 5000 | 0.0 | **3.11 (± 0.05)** | 3.11 (± 0.05) | 0.94 (± 0.40) | **0.93 (± 0.43)** | 15.72 (± 1.16) | 1.31 (± 0.71) |
| | | 0.3 | 3.09 (± 0.05) | **3.02 (± 0.05)** | 0.98 (± 0.30) | **0.97 (± 0.31)** | 15.70 (± 1.29) | 5.98 (± 1.21) |
| | | 0.6 | 3.08 (± 0.05) | **2.75 (± 0.04)** | 1.08 (± 0.33) | **1.05 (± 0.32)** | 15.72 (± 1.37) | 11.95 (± 1.01) |
| | | 0.9 | 3.08 (± 0.06) | **2.15 (± 0.03)** | 1.17 (± 0.36) | **1.12 (± 0.33)** | 15.75 (± 1.50) | 17.89 (± 0.70) |
| $\mathbf{A}_6(t)$ | 2000 | 0.0 | **4.40 (± 0.10)** | 4.42 (± 0.10) | 1.61 (± 0.74) | **1.61 (± 0.76)** | 23.98 (± 2.01) | 1.83 (± 1.08) |
| | | 0.3 | 4.40 (± 0.11) | **4.35 (± 0.12)** | 1.79 (± 0.70) | **1.68 (± 0.72)** | 24.06 (± 2.15) | 8.40 (± 2.57) |
| | | 0.6 | 4.40 (± 0.11) | **4.08 (± 0.11)** | 1.94 (± 0.76) | **1.81 (± 0.79)** | 23.96 (± 2.56) | 17.82 (± 2.12) |
| | | 0.9 | 4.53 (± 0.13) | **3.50 (± 0.11)** | 2.08 (± 0.84) | **1.90 (± 0.87)** | 31.74 (± 2.88) | 27.29 (± 1.34) |
| | 5000 | 0.0 | **2.66 (± 0.05)** | 2.66 (± 0.05) | 0.93 (± 0.34) | **0.91 (± 0.36)** | 13.99 (± 1.23) | 1.23 (± 0.78) |
| | | 0.3 | 2.64 (± 0.05) | **2.57 (± 0.05)** | 0.96 (± 0.31) | **0.94 (± 0.31)** | 13.96 (± 1.28) | 4.60 (± 1.14) |
| | | 0.6 | 2.64 (± 0.05) | **2.32 (± 0.05)** | 1.06 (± 0.34) | **1.02 (± 0.32)** | 14.00 (± 1.38) | 9.15 (± 0.92) |
| | | 0.9 | 2.76 (± 0.06) | **1.66 (± 0.04)** | 1.16 (± 0.37) | **1.09 (± 0.34)** | 19.41 (± 1.51) | 14.58 (± 0.51) |

Table 7: MAE errors ($\times 10^{-2}$) for Case 2(c) - $\sigma_2^2(t) = 1 + 0.25\cos(2\pi t) + 0.1\sin(4\pi t)$.

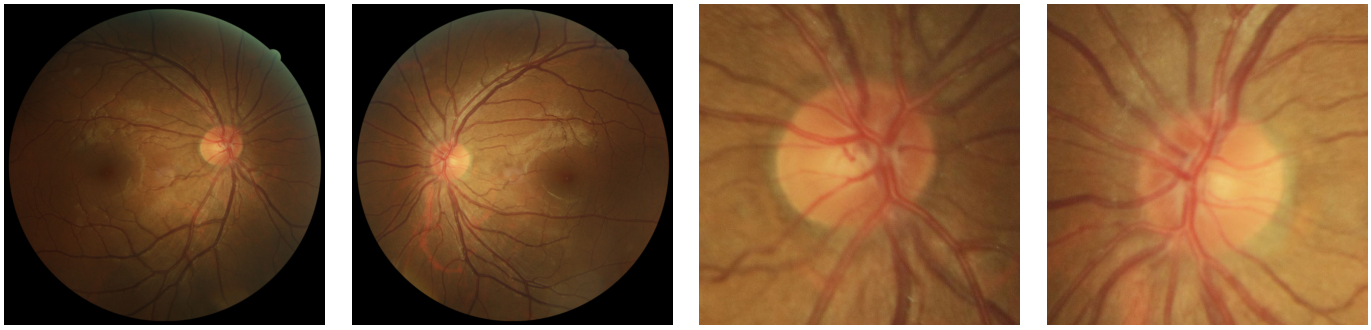
| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^{\dagger}(t)$ | $\hat{\mathbf{A}}^{*}(t)$ | $\hat{\boldsymbol{\beta}}^{\dagger}$ | $\hat{\boldsymbol{\beta}}^{*}$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 2000 | 0.0 | **3.60 (± 0.09)** | 3.60 (± 0.09) | **1.51 (± 0.68)** | 1.52 (± 0.68) | 23.71 (± 2.11) | 1.91 (± 1.38) |
| | | 0.3 | 3.59 (± 0.10) | **3.50 (± 0.10)** | 1.70 (± 0.71) | **1.67 (± 0.70)** | 23.57 (± 1.96) | 6.71 (± 2.51) |
| | | 0.6 | 3.59 (± 0.10) | **3.20 (± 0.09)** | 1.87 (± 0.77) | **1.81 (± 0.76)** | 23.57 (± 2.36) | 13.85 (± 1.93) |
| | | 0.9 | 3.59 (± 0.11) | **2.53 (± 0.08)** | 2.02 (± 0.82) | **1.93 (± 0.78)** | 23.61 (± 2.78) | 21.00 (± 1.23) |
| | 5000 | 0.0 | 2.32 (± 0.06) | **2.31 (± 0.06)** | 0.92 (± 0.37) | **0.89 (± 0.36)** | 8.74 (± 1.05) | 1.06 (± 0.74) |
| | | 0.3 | 2.29 (± 0.06) | **2.23 (± 0.06)** | 0.94 (± 0.30) | **0.93 (± 0.29)** | 8.81 (± 1.05) | 3.82 (± 1.32) |
| | | 0.6 | 2.28 (± 0.06) | **2.05 (± 0.05)** | 1.04 (± 0.33) | **1.02 (± 0.31)** | 8.95 (± 1.24) | 7.32 (± 1.06) |
| | | 0.9 | 2.28 (± 0.06) | **1.66 (± 0.04)** | 1.14 (± 0.37) | **1.10 (± 0.33)** | 9.11 (± 1.49) | 10.76 (± 0.63) |
| $\mathbf{A}_2(t)$ | 2000 | 0.0 | **3.56 (± 0.10)** | 3.57 (± 0.10) | **1.48 (± 0.73)** | 1.49 (± 0.72) | 23.95 (± 2.07) | 1.66 (± 1.32) |
| | | 0.3 | 3.56 (± 0.10) | **3.47 (± 0.10)** | 1.73 (± 0.69) | **1.70 (± 0.69)** | 23.82 (± 1.90) | 6.48 (± 2.41) |
| | | 0.6 | 3.56 (± 0.10) | **3.17 (± 0.09)** | 1.89 (± 0.75) | **1.83 (± 0.75)** | 23.85 (± 2.25) | 13.54 (± 1.96) |
| | | 0.9 | 3.56 (± 0.11) | **2.46 (± 0.08)** | 2.05 (± 0.80) | **1.95 (± 0.80)** | 23.91 (± 2.65) | 20.66 (± 1.36) |
| | 5000 | 0.0 | 2.28 (± 0.06) | **2.27 (± 0.05)** | 0.92 (± 0.38) | **0.90 (± 0.37)** | 10.41 (± 1.04) | 1.15 (± 0.71) |
| | | 0.3 | 2.26 (± 0.05) | **2.20 (± 0.04)** | 0.92 (± 0.31) | **0.92 (± 0.28)** | 10.45 (± 1.07) | 3.76 (± 1.24) |
| | | 0.6 | 2.26 (± 0.05) | **2.00 (± 0.04)** | 1.02 (± 0.34) | **1.00 (± 0.31)** | 10.56 (± 1.24) | 7.25 (± 0.99) |
| | | 0.9 | 2.25 (± 0.05) | **1.55 (± 0.03)** | 1.12 (± 0.36) | **1.08 (± 0.32)** | 10.69 (± 1.48) | 10.69 (± 0.62) |
| $\mathbf{A}_3(t)$ | 2000 | 0.0 | **4.63 (± 0.11)** | 4.66 (± 0.11) | 1.54 (± 0.71) | **1.52 (± 0.71)** | 23.47 (± 2.06) | 1.91 (± 1.56) |
| | | 0.3 | 4.63 (± 0.11) | **4.59 (± 0.10)** | 1.75 (± 0.81) | **1.73 (± 0.80)** | 23.53 (± 2.03) | 9.63 (± 2.37) |
| | | 0.6 | 4.69 (± 0.13) | **4.32 (± 0.11)** | 1.93 (± 0.87) | **1.86 (± 0.88)** | 32.34 (± 2.26) | 18.91 (± 1.69) |
| | | 0.9 | 4.69 (± 0.14) | **3.70 (± 0.10)** | 2.07 (± 0.91) | **1.95 (± 0.91)** | 32.29 (± 2.68) | 28.85 (± 1.03) |
| | 5000 | 0.0 | **2.72 (± 0.05)** | 2.72 (± 0.05) | 0.94 (± 0.33) | **0.92 (± 0.31)** | 14.38 (± 1.22) | 1.07 (± 0.70) |
| | | 0.3 | 2.70 (± 0.06) | **2.63 (± 0.05)** | 0.95 (± 0.36) | **0.94 (± 0.34)** | 14.32 (± 1.21) | 5.05 (± 1.21) |
| | | 0.6 | 2.70 (± 0.06) | **2.38 (± 0.05)** | 1.05 (± 0.38) | **1.02 (± 0.36)** | 14.40 (± 1.34) | 9.74 (± 0.98) |
| | | 0.9 | 2.80 (± 0.06) | **1.69 (± 0.04)** | 1.16 (± 0.40) | **1.09 (± 0.35)** | 20.05 (± 1.52) | 14.99 (± 0.61) |
| $\mathbf{A}_4(t)$ | 2000 | 0.0 | **3.33 (± 0.09)** | 3.33 (± 0.09) | **1.48 (± 0.70)** | 1.49 (± 0.69) | 26.11 (± 2.14) | 1.81 (± 1.44) |
| | | 0.3 | 3.32 (± 0.09) | **3.23 (± 0.08)** | 1.69 (± 0.71) | **1.65 (± 0.71)** | 25.99 (± 2.02) | 5.93 (± 2.42) |
| | | 0.6 | 3.32 (± 0.09) | **2.89 (± 0.08)** | 1.86 (± 0.77) | **1.79 (± 0.78)** | 25.97 (± 2.46) | 12.28 (± 1.82) |
| | | 0.9 | 3.33 (± 0.11) | **2.06 (± 0.07)** | 2.01 (± 0.82) | **1.91 (± 0.80)** | 25.98 (± 2.93) | 18.68 (± 1.01) |
| | 5000 | 0.0 | **1.93 (± 0.04)** | 1.93 (± 0.04) | 0.92 (± 0.34) | **0.89 (± 0.33)** | 10.46 (± 1.17) | 1.09 (± 0.70) |
| | | 0.3 | 1.90 (± 0.05) | **1.83 (± 0.05)** | 0.93 (± 0.31) | **0.92 (± 0.29)** | 10.47 (± 1.14) | 3.03 (± 1.25) |
| | | 0.6 | 1.90 (± 0.06) | **1.60 (± 0.05)** | 1.03 (± 0.34) | **1.01 (± 0.31)** | 10.59 (± 1.30) | 5.68 (± 0.96) |
| | | 0.9 | 1.90 (± 0.06) | **1.04 (± 0.03)** | 1.13 (± 0.37) | **1.09 (± 0.33)** | 10.72 (± 1.54) | 8.25 (± 0.51) |
| $\mathbf{A}_5(t)$ | 2000 | 0.0 | **5.45 (± 0.13)** | 5.49 (± 0.13) | 1.68 (± 0.66) | **1.67 (± 0.70)** | 26.59 (± 1.30) | 2.02 (± 1.32) |
| | | 0.3 | 5.47 (± 0.16) | **5.44 (± 0.16)** | 1.95 (± 0.88) | **1.86 (± 0.92)** | 26.57 (± 1.45) | 12.05 (± 2.44) |
| | | 0.6 | 5.47 (± 0.17) | **5.20 (± 0.16)** | 2.09 (± 0.94) | **1.98 (± 1.00)** | 26.61 (± 1.65) | 24.82 (± 2.24) |
| | | 0.9 | 5.48 (± 0.18) | **4.74 (± 0.17)** | 2.22 (± 0.97) | **2.09 (± 1.04)** | 26.69 (± 1.91) | 37.63 (± 2.03) |
| | 5000 | 0.0 | **3.16 (± 0.05)** | 3.16 (± 0.05) | 0.95 (± 0.43) | **0.94 (± 0.41)** | 16.81 (± 1.06) | 1.23 (± 0.66) |
| | | 0.3 | 3.14 (± 0.05) | **3.06 (± 0.05)** | 0.99 (± 0.31) | **0.98 (± 0.28)** | 16.77 (± 1.09) | 6.14 (± 1.25) |
| | | 0.6 | 3.13 (± 0.06) | **2.80 (± 0.04)** | 1.09 (± 0.34) | **1.06 (± 0.30)** | 16.83 (± 1.21) | 12.18 (± 1.05) |
| | | 0.9 | 3.13 (± 0.06) | **2.18 (± 0.03)** | 1.18 (± 0.36) | **1.13 (± 0.32)** | 16.90 (± 1.38) | 18.17 (± 0.74) |
| $\mathbf{A}_6(t)$ | 2000 | 0.0 | **4.46 (± 0.10)** | 4.48 (± 0.11) | **1.58 (± 0.78)** | 1.59 (± 0.76) | 25.20 (± 1.96) | 1.81 (± 1.11) |
| | | 0.3 | 4.46 (± 0.11) | **4.41 (± 0.12)** | 1.76 (± 0.73) | **1.72 (± 0.73)** | 25.20 (± 2.05) | 8.76 (± 2.61) |
| | | 0.6 | 4.46 (± 0.11) | **4.14 (± 0.11)** | 1.92 (± 0.79) | **1.85 (± 0.80)** | 25.16 (± 2.46) | 18.40 (± 2.15) |
| | | 0.9 | 4.60 (± 0.13) | **3.57 (± 0.12)** | 2.05 (± 0.88) | **1.94 (± 0.87)** | 32.94 (± 2.77) | 27.95 (± 1.33) |
| | 5000 | 0.0 | **2.68 (± 0.05)** | 2.68 (± 0.05) | 0.93 (± 0.36) | **0.90 (± 0.34)** | 14.65 (± 1.22) | 1.15 (± 0.76) |
| | | 0.3 | 2.66 (± 0.05) | **2.59 (± 0.05)** | 0.96 (± 0.32) | **0.94 (± 0.31)** | 14.62 (± 1.25) | 4.79 (± 1.21) |
| | | 0.6 | 2.65 (± 0.05) | **2.33 (± 0.05)** | 1.06 (± 0.35) | **1.02 (± 0.32)** | 14.69 (± 1.37) | 9.41 (± 0.98) |
| | | 0.9 | 2.79 (± 0.06) | **1.67 (± 0.04)** | 1.17 (± 0.38) | **1.09 (± 0.34)** | 20.11 (± 1.53) | 14.86 (± 0.55) |

Table 8: MAE errors ($\times 10^{-2}$) for Case 2(d) - $\sigma_3^2(t) = 0.5 + 0.5\sin^2(\pi t)$.

| function | $n$ | $\rho$ | MIAE | | MAE | | MIAE | MAE |
|---|---|---|---|---|---|---|---|---|
| | | | $\hat{\mathbf{A}}^\dagger(t)$ | $\hat{\mathbf{A}}^*(t)$ | $\hat{\boldsymbol{\beta}}^\dagger$ | $\hat{\boldsymbol{\beta}}^*$ | $\hat{\sigma}^2(t)$ | $\hat{\rho}$ |
| $\mathbf{A}_1(t)$ | 2000 | 0.0 | **3.03 (± 0.09)** | 3.04 (± 0.09) | 1.38 (± 0.59) | **1.29 (± 0.55)** | 15.01 (± 1.82) | 2.13 (± 1.48) |
| | | 0.3 | 3.02 (± 0.08) | **2.96 (± 0.08)** | 1.48 (± 0.61) | **1.41 (± 0.60)** | 14.82 (± 1.76) | 6.41 (± 2.66) |
| | | 0.6 | 3.03 (± 0.09) | **2.72 (± 0.08)** | 1.61 (± 0.68) | **1.53 (± 0.64)** | 14.76 (± 2.15) | 13.17 (± 2.18) |
| | | 0.9 | 3.03 (± 0.09) | **2.19 (± 0.07)** | 1.74 (± 0.74) | **1.64 (± 0.65)** | 14.73 (± 2.57) | 20.00 (± 1.39) |
| | 5000 | 0.0 | **2.03 (± 0.05)** | 2.04 (± 0.05) | 0.78 (± 0.30) | **0.77 (± 0.32)** | 3.64 (± 0.97) | 1.21 (± 0.88) |
| | | 0.3 | 2.00 (± 0.05) | **1.98 (± 0.05)** | 0.82 (± 0.26) | **0.79 (± 0.25)** | 3.57 (± 0.92) | 3.76 (± 1.33) |
| | | 0.6 | 2.00 (± 0.05) | **1.84 (± 0.05)** | 0.91 (± 0.29) | **0.86 (± 0.27)** | 3.67 (± 0.98) | 7.37 (± 1.07) |
| | | 0.9 | 2.00 (± 0.05) | **1.57 (± 0.03)** | 0.99 (± 0.31) | **0.93 (± 0.29)** | 3.81 (± 1.07) | 10.93 (± 0.66) |
| $\mathbf{A}_2(t)$ | 2000 | 0.0 | **2.98 (± 0.09)** | 3.00 (± 0.09) | 1.33 (± 0.62) | **1.24 (± 0.61)** | 15.17 (± 1.84) | 1.86 (± 1.54) |
| | | 0.3 | 2.98 (± 0.08) | **2.92 (± 0.09)** | 1.50 (± 0.59) | **1.43 (± 0.59)** | 15.01 (± 1.79) | 6.22 (± 2.33) |
| | | 0.6 | 2.98 (± 0.09) | **2.68 (± 0.08)** | 1.64 (± 0.64) | **1.54 (± 0.65)** | 14.96 (± 2.16) | 12.81 (± 2.16) |
| | | 0.9 | 2.98 (± 0.09) | **2.11 (± 0.07)** | 1.78 (± 0.70) | **1.65 (± 0.66)** | 14.92 (± 2.56) | 19.60 (± 1.53) |
| | 5000 | 0.0 | **1.98 (± 0.04)** | 1.99 (± 0.04) | 0.78 (± 0.31) | **0.77 (± 0.33)** | 4.30 (± 0.79) | 1.32 (± 0.69) |
| | | 0.3 | 1.96 (± 0.04) | **1.93 (± 0.04)** | 0.81 (± 0.25) | **0.76 (± 0.26)** | 4.23 (± 0.81) | 3.70 (± 1.30) |
| | | 0.6 | 1.95 (± 0.04) | **1.78 (± 0.04)** | 0.90 (± 0.27) | **0.83 (± 0.28)** | 4.30 (± 0.88) | 7.30 (± 1.05) |
| | | 0.9 | 1.95 (± 0.05) | **1.44 (± 0.03)** | 0.98 (± 0.30) | **0.89 (± 0.29)** | 4.41 (± 0.98) | 10.86 (± 0.69) |
| $\mathbf{A}_3(t)$ | 2000 | 0.0 | **4.00 (± 0.10)** | 4.01 (± 0.10) | 1.37 (± 0.61) | **1.34 (± 0.58)** | 21.36 (± 1.46) | 2.23 (± 1.52) |
| | | 0.3 | 4.00 (± 0.09) | **3.94 (± 0.09)** | 1.55 (± 0.65) | **1.50 (± 0.71)** | 21.30 (± 1.65) | 9.07 (± 2.43) |
| | | 0.6 | 4.00 (± 0.10) | **3.70 (± 0.08)** | 1.68 (± 0.71) | **1.60 (± 0.78)** | 21.23 (± 1.96) | 18.91 (± 1.81) |
| | | 0.9 | 4.00 (± 0.11) | **3.21 (± 0.08)** | 1.79 (± 0.77) | **1.67 (± 0.80)** | 21.17 (± 2.27) | 28.79 (± 1.11) |
| | 5000 | 0.0 | **2.38 (± 0.05)** | 2.38 (± 0.05) | 0.80 (± 0.25) | **0.79 (± 0.29)** | 9.03 (± 0.96) | 1.14 (± 0.74) |
| | | 0.3 | 2.37 (± 0.05) | **2.31 (± 0.05)** | 0.83 (± 0.29) | **0.79 (± 0.31)** | 8.95 (± 0.96) | 5.00 (± 1.20) |
| | | 0.6 | 2.40 (± 0.05) | **2.08 (± 0.04)** | 0.92 (± 0.31) | **0.86 (± 0.32)** | 13.69 (± 1.03) | 9.88 (± 0.93) |
| | | 0.9 | 2.40 (± 0.05) | **1.48 (± 0.03)** | 1.00 (± 0.32) | **0.91 (± 0.31)** | 13.71 (± 1.14) | 14.73 (± 0.59) |
| $\mathbf{A}_4(t)$ | 2000 | 0.0 | **2.75 (± 0.08)** | 2.75 (± 0.08) | 1.34 (± 0.60) | **1.24 (± 0.56)** | 17.41 (± 1.85) | 1.98 (± 1.59) |
| | | 0.3 | 2.74 (± 0.07) | **2.66 (± 0.07)** | 1.46 (± 0.61) | **1.40 (± 0.62)** | 17.24 (± 1.81) | 5.47 (± 2.38) |
| | | 0.6 | 2.75 (± 0.08) | **2.38 (± 0.06)** | 1.60 (± 0.68) | **1.52 (± 0.66)** | 17.19 (± 2.18) | 11.14 (± 2.00) |
| | | 0.9 | 2.75 (± 0.09) | **1.69 (± 0.05)** | 1.74 (± 0.73) | **1.62 (± 0.67)** | 17.15 (± 2.60) | 16.99 (± 1.07) |
| | 5000 | 0.0 | 1.62 (± 0.04) | **1.62 (± 0.04)** | 0.78 (± 0.28) | **0.77 (± 0.30)** | 6.40 (± 1.03) | 1.20 (± 0.78) |
| | | 0.3 | 1.60 (± 0.05) | **1.55 (± 0.04)** | 0.81 (± 0.25) | **0.77 (± 0.26)** | 6.33 (± 0.98) | 2.74 (± 1.19) |
| | | 0.6 | 1.60 (± 0.05) | **1.36 (± 0.04)** | 0.90 (± 0.28) | **0.85 (± 0.28)** | 6.39 (± 1.06) | 5.22 (± 0.96) |
| | | 0.9 | 1.60 (± 0.05) | **0.91 (± 0.03)** | 0.98 (± 0.31) | **0.92 (± 0.29)** | 6.46 (± 1.21) | 7.66 (± 0.51) |
| $\mathbf{A}_5(t)$ | 2000 | 0.0 | **4.80 (± 0.11)** | 4.84 (± 0.11) | 1.49 (± 0.58) | **1.44 (± 0.58)** | 15.36 (± 1.29) | 2.16 (± 1.55) |
| | | 0.3 | 4.81 (± 0.13) | **4.79 (± 0.13)** | 1.71 (± 0.73) | **1.59 (± 0.81)** | 15.14 (± 1.36) | 12.72 (± 2.63) |
| | | 0.6 | 4.81 (± 0.14) | **4.61 (± 0.14)** | 1.83 (± 0.77) | **1.71 (± 0.87)** | 15.17 (± 1.47) | 26.05 (± 2.49) |
| | | 0.9 | 4.81 (± 0.15) | **4.26 (± 0.15)** | 1.93 (± 0.81) | **1.80 (± 0.91)** | 15.22 (± 1.61) | 39.43 (± 2.29) |
| | 5000 | 0.0 | **2.77 (± 0.05)** | 2.77 (± 0.05) | 0.82 (± 0.33) | **0.80 (± 0.38)** | 10.16 (± 0.87) | 1.34 (± 0.74) |
| | | 0.3 | 2.75 (± 0.05) | **2.69 (± 0.04)** | 0.87 (± 0.24) | **0.83 (± 0.26)** | 10.10 (± 0.96) | 6.26 (± 1.32) |
| | | 0.6 | 2.74 (± 0.05) | **2.48 (± 0.04)** | 0.96 (± 0.27) | **0.90 (± 0.27)** | 10.12 (± 1.02) | 12.57 (± 1.10) |
| | | 0.9 | 2.74 (± 0.05) | **1.99 (± 0.03)** | 1.04 (± 0.29) | **0.96 (± 0.28)** | 10.15 (± 1.13) | 18.86 (± 0.77) |
| $\mathbf{A}_6(t)$ | 2000 | 0.0 | **3.93 (± 0.09)** | 3.94 (± 0.09) | 1.41 (± 0.67) | **1.40 (± 0.68)** | 15.22 (± 1.60) | 1.94 (± 1.20) |
| | | 0.3 | 3.93 (± 0.10) | **3.88 (± 0.10)** | 1.54 (± 0.63) | **1.48 (± 0.61)** | 15.11 (± 1.83) | 8.87 (± 2.71) |
| | | 0.6 | 3.96 (± 0.10) | **3.65 (± 0.10)** | 1.65 (± 0.72) | **1.57 (± 0.70)** | 21.89 (± 2.07) | 18.05 (± 2.10) |
| | | 0.9 | 3.95 (± 0.11) | **3.11 (± 0.09)** | 1.77 (± 0.77) | **1.65 (± 0.73)** | 21.83 (± 2.36) | 27.72 (± 1.50) |
| | 5000 | 0.0 | 2.37 (± 0.04) | **2.36 (± 0.04)** | **0.79 (± 0.29)** | 0.79 (± 0.30) | 9.31 (± 0.96) | 1.27 (± 0.78) |
| | | 0.3 | 2.35 (± 0.05) | **2.29 (± 0.04)** | 0.84 (± 0.26) | **0.80 (± 0.28)** | 9.26 (± 0.96) | 4.70 (± 1.24) |
| | | 0.6 | 2.35 (± 0.05) | **2.09 (± 0.04)** | 0.93 (± 0.28) | **0.86 (± 0.29)** | 9.29 (± 1.02) | 9.40 (± 1.02) |
| | | 0.9 | 2.40 (± 0.05) | **1.49 (± 0.03)** | 1.01 (± 0.30) | **0.92 (± 0.30)** | 13.77 (± 1.15) | 14.57 (± 0.58) |

## Empirical Experiments

**Dataset Information.** The glaucoma real-world appraisal progression ensemble (GRAPE) is a longitudinal multimodal dataset collected for glaucoma management. It contains fundus images together with clinical information collected during follow-up. For each image visit, two image types data are collected: a color fundus photograph (CFP), an optic-disc-centered region-of-interest (ROI) image. An example is given in Figure 5.



(a) CFP, OD. (b) CFP, OS. (c) ROI, OD. (d) ROI, OS.

Figure 5: Original CFP and ROI images from an representative paired visit.

The intraocular pressure (IOP), age, gender, and visual-field (VF) measurements are also included in the dataset. Since elevated IOP is an important biomarker and modifiable risk factor for glaucoma progression, our objective is to quantify how the association between fundus images and IOP varies with age and to evaluate the predictive value of the image and clinical covariates.

**Preprocessing.** The distributed image subset contains 631 eye-level visits with matched CFP and ROI images. We paired oculus dexter (OD) and oculus sinister (OS) records from the same subject at the same follow-up time, obtaining 276 paired visits. After excluding a paired visit whenever the IOP of either eye exceeded 35 mmHg, the primary analysis retained 273 paired visits, corresponding to 546 eye-level observations for each image representation. All images were resized to $192 \times 192 \times 3$, and OS images were horizontally flipped to align with the OD images.

Let $i = 1, \ldots, 273$ index paired visits and $j \in \{1, 2\}$ index the OD and OS eyes, respectively. For either the CFP or ROI analysis, let $\mathcal{X}_{ij} \in \mathbb{R}^{192\times192\times3}$ denote the image tensor and let $y_{ij}$ denote the corresponding IOP. The index variable $t_i$ is age, min-max normalized to $[0, 1]$. The pair-level covariate vector $\mathbf{z}_i \in \mathbb{R}^{60}$ contains gender and the paired-eye averages of 59 non-blind-spot VF measurements; the continuous VF covariates and IOP outcomes are standardized before model fitting. We then apply the paired-eye VCTR model

$$y_{ij} = \langle \mathcal{X}_{ij}, \mathcal{A}(t_i) \rangle + \mathbf{z}_i^\top \boldsymbol{\beta} + \epsilon_{ij}, \quad \text{where} \quad \begin{bmatrix} \epsilon_{i1} \\ \epsilon_{i2} \end{bmatrix} \sim \mathbf{N}\left( \mathbf{0},\ \sigma^2(t_i) \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right),$$

and $\mathcal{A}(t)$ captures the age-dependent image effect and $\boldsymbol{\beta}$ contains the constant effects of the clinical covariates.

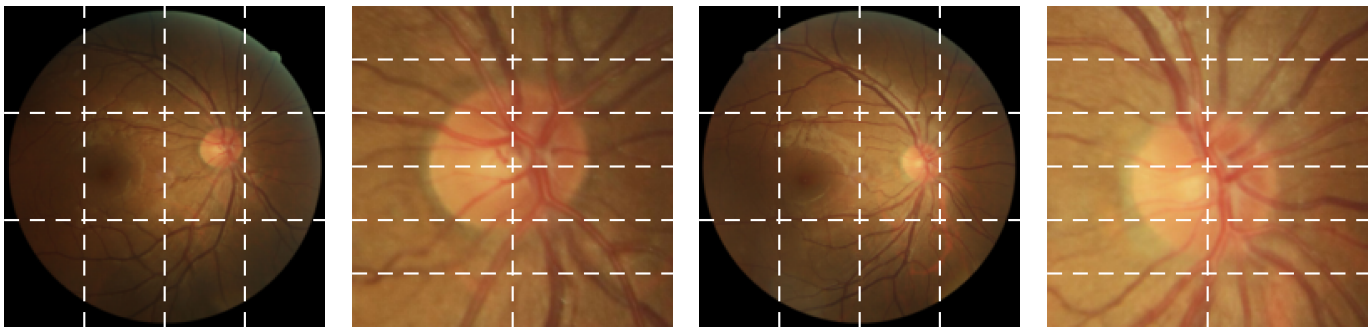
**Choosing Hyperparameter.** We select four hyperparameters separately for CFP and ROI: the partition grid $S^{(1)} \times S^{(2)} \times S^{(3)}$, the blockwise CP rank $R$, the bandwidth $h$ for estimating the varying image effect, and the bandwidth $\bar{h}$ for estimating the variance function. The candidate feature set contains 17 partition grids and 56 available $(S, R)$ combinations. For each feature configuration, we consider 32 values of $h$ ranging from 0.18 to 8.00 and 15 values of $\bar{h}$ ranging from 0.20 to 4.00. This gives $56 \times 32 \times 15 = 26{,}880$ candidates for each image type and 53,760 candidates in total. Hyperparameters are selected using 5-fold cross-validation.

Table 9 reports, for each $(S, R)$ configuration, the smallest cross-validated RMSE over all 480 combinations of $(h, \bar{h})$. The selected configurations are $S^{(1)} \times S^{(2)} \times S^{(3)} = 3 \times 4 \times 1$, $R = 1$, $h = 1.80$, and $\bar{h} = 0.25$ for CFP, and $6 \times 2 \times 1$, $R = 1$, $h = 0.85$, and $\bar{h} = 0.30$ for ROI. Their standardized-scale RMSE values are 0.9890 and 0.9620.

Table 9: Minimum RMSE by tensor partition and CP rank.

| $S^{(1)} \times S^{(2)} \times S^{(3)}$ | CFP | | | | ROI | | | |
|---|---|---|---|---|---|---|---|---|
| | $R = 1$ | $R = 2$ | $R = 3$ | $R = 4$ | $R = 1$ | $R = 2$ | $R = 3$ | $R = 4$ |
| $2 \times 2 \times 1$ | 0.9958 | 0.9993 | 1.0068 | 1.0125 | 0.9793 | 0.9746 | 0.9942 | 0.9964 |
| $2 \times 3 \times 1$ | 1.0032 | 1.0213 | 1.0207 | – | 0.9849 | 0.9942 | 1.0219 | – |
| $2 \times 4 \times 1$ | 1.0096 | 1.0219 | 1.0212 | – | 0.9924 | 1.0089 | 1.0370 | – |
| $2 \times 6 \times 1$ | 1.0204 | 1.0369 | 1.1059 | – | 1.0093 | 1.0410 | 1.0448 | – |
| $3 \times 2 \times 1$ | 0.9931 | 0.9993 | 1.0247 | – | 0.9742 | 0.9936 | 1.0190 | – |
| $3 \times 3 \times 1$ | 0.9964 | 1.0261 | 1.0383 | 1.0440 | 0.9782 | 0.9978 | 1.0563 | 1.0698 |
| $3 \times 4 \times 1$ | **0.9890** | 1.0025 | 1.0547 | – | 1.0054 | 1.0521 | 1.0905 | – |
| $3 \times 6 \times 1$ | 1.0087 | 1.0596 | 1.1367 | – | 1.0112 | 1.0890 | 1.1446 | – |
| $4 \times 2 \times 1$ | 1.0055 | 1.0135 | 1.0326 | – | 0.9766 | 0.9885 | 1.0242 | – |
| $4 \times 3 \times 1$ | 1.0054 | 1.0626 | 1.1094 | – | 0.9823 | 1.0246 | 1.0476 | – |
| $4 \times 4 \times 1$ | 1.0131 | 1.0382 | 1.1292 | 1.1915 | 1.0023 | 1.0397 | 1.1207 | 1.1167 |
| $4 \times 6 \times 1$ | 1.0363 | 1.1582 | 1.2898 | – | 0.9768 | 1.0598 | 1.1639 | – |
| $6 \times 2 \times 1$ | 1.0160 | 1.0358 | 1.0586 | – | **0.9620** | 0.9792 | 1.0512 | – |
| $6 \times 3 \times 1$ | 1.0231 | 1.0709 | 1.1856 | – | 0.9904 | 1.0589 | 1.1264 | – |
| $6 \times 4 \times 1$ | 1.0231 | 1.0666 | 1.1431 | – | 1.0174 | 1.1024 | 1.2372 | – |
| $6 \times 6 \times 1$ | 1.0701 | 1.3510 | 1.7346 | 2.2838 | 1.0444 | 1.1539 | 1.3007 | 1.7721 |
| $8 \times 8 \times 1$ | 1.1220 | 1.7289 | 3.4191 | 5.0691 | 1.1446 | 1.4350 | 3.5168 | 4.7495 |

Figure 6 shows an image example under our chosen hyperparameter.



(a) CFP, OD. (b) ROI, OD. (c) CFP, OS. (d) ROI, OS.

Figure 6: Representative paired-eye GRAPE images with partition $3 \times 4 \times 1$ for CFP and $6 \times 2 \times 1$ for ROI.

**ROI Model Interpretation.** For the VCTR model where $\mathcal{X}_{ij}$ represents ROI image, it uses the configuration $S^{(1)} \times S^{(2)} \times S^{(3)} = 6 \times 2 \times 1$, $R = 1$, $h = 0.85$, and $\bar{h} = 0.30$. Figure 7 presents the estimated coefficient functions $\hat{a}_{1s}(t)$. Figure 8 shows the estimate noise variance function $\hat{\sigma}(t)$. And Table 10 shows the full estimates of VFs.

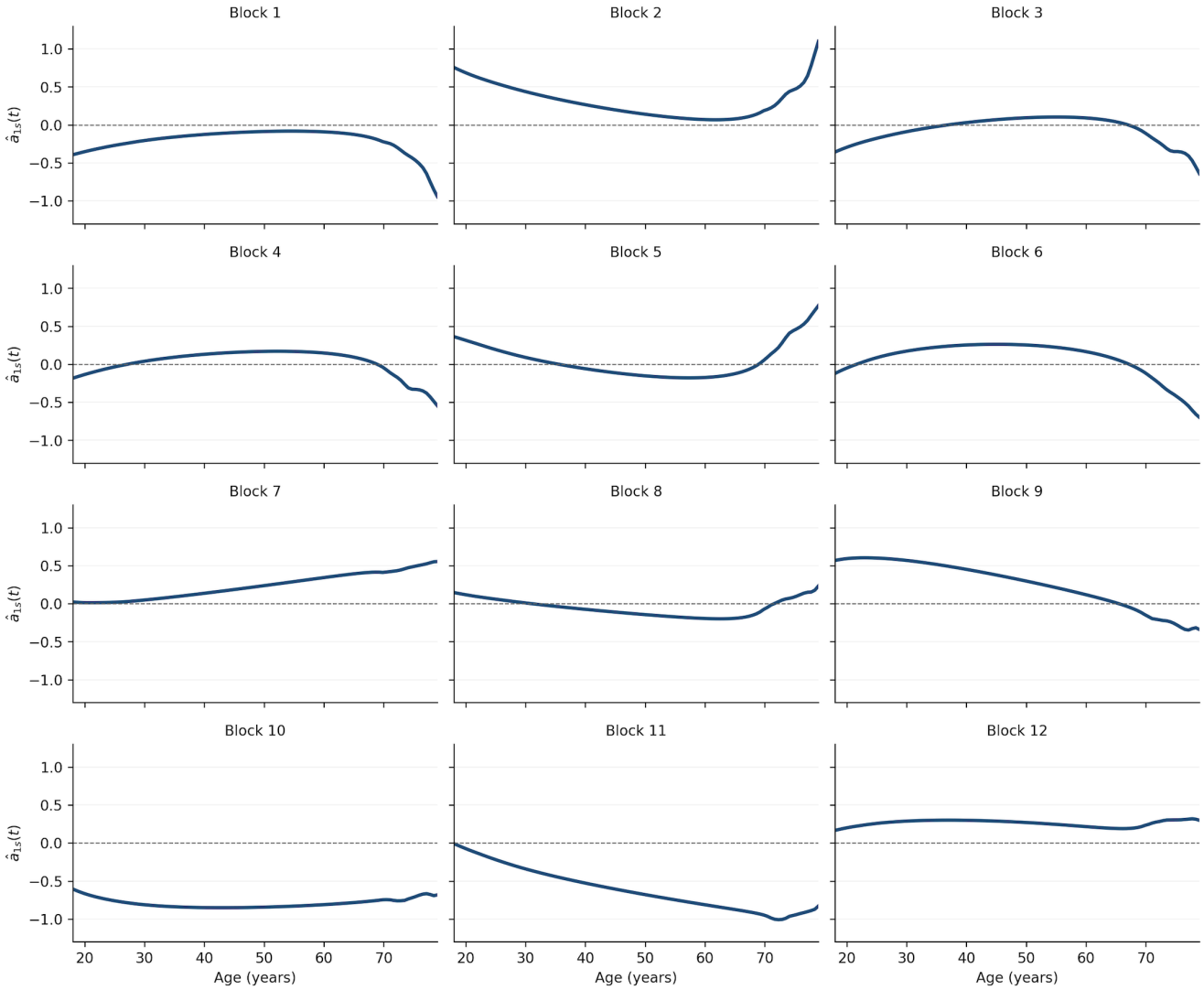
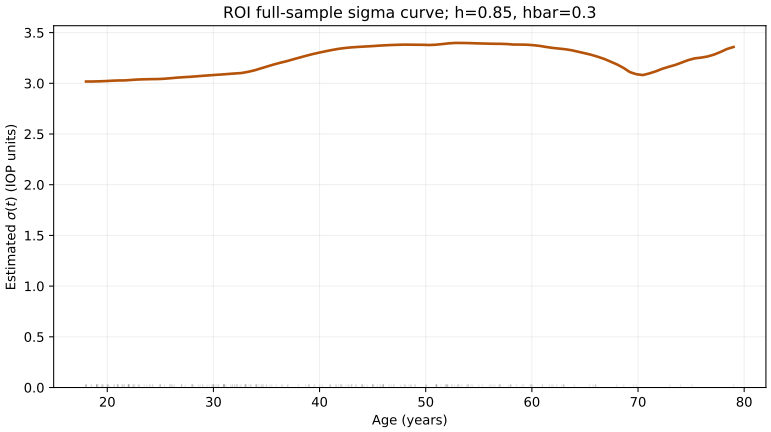
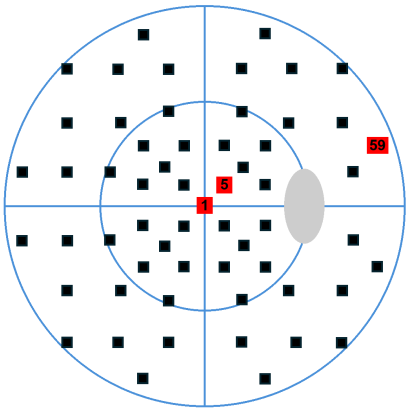



Figure 7: Estimated coefficient functions $\hat{\alpha}(t)$




(a) Estimated $\sigma(t)$.




(b) Selected VF locations.

Figure 8: Estimate $\hat{\sigma}(t)$ and significant VF locations.

Table 10: All scalar coefficients in ROI VCTR model based on a $B = 2000$ bootstrap.

| Covariate | Estimate | Bootstrap SE | 95% CI | $p$-value |
|---|---|---|---|---|
| Gender | 0.3636 | 0.7484 | [−1.0519, 1.8401] | 0.543 |
| **VF00** | **−4.7368** | **1.8050** | **[−7.7807, −0.7137]** | **0.023** |
| VF01 | 2.4040 | 1.6152 | [−1.0620, 5.2806] | 0.148 |
| VF02 | 1.2436 | 2.0067 | [−2.9930, 4.9205] | 0.522 |
| VF03 | 3.8123 | 2.2362 | [−0.3617, 8.3195] | 0.070 |
| **VF04** | **4.6147** | **2.0020** | **[0.7861, 8.5312]** | **0.019** |
| VF05 | −3.2855 | 2.2925 | [−8.2329, 1.0029] | 0.112 |
| VF06 | −1.2673 | 2.3895 | [−6.5481, 2.9152] | 0.447 |
| VF07 | 1.9111 | 2.8446 | [−3.6889, 7.6151] | 0.570 |
| VF08 | −2.1027 | 2.1817 | [−6.6136, 1.7527] | 0.295 |
| VF09 | 0.6285 | 2.1515 | [−3.4967, 5.0914] | 0.689 |
| VF10 | −0.1016 | 1.7733 | [−3.7164, 3.2306] | 0.906 |
| VF11 | 0.9169 | 1.7497 | [−2.4376, 4.6234] | 0.502 |
| VF12 | −0.7978 | 2.2431 | [−4.4937, 4.3516] | 0.886 |
| VF13 | 3.9601 | 2.2563 | [−0.8856, 7.9618] | 0.119 |
| VF14 | 0.4009 | 2.0320 | [−1.8102, 6.2806] | 0.758 |
| VF15 | −1.7483 | 2.6266 | [−7.2105, 3.0866] | 0.454 |
| VF16 | 4.7956 | 2.7778 | [−0.0749, 10.6414] | 0.053 |
| VF17 | −3.2574 | 2.9900 | [−9.2973, 2.1048] | 0.286 |
| VF18 | −2.7720 | 2.3446 | [−7.2509, 2.0119] | 0.253 |
| VF19 | 2.8077 | 2.2909 | [−1.8514, 6.9746] | 0.252 |
| VF20 | −0.5433 | 2.3672 | [−5.0245, 4.2213] | 0.826 |
| VF22 | 1.8373 | 1.8401 | [−1.7735, 5.5462] | 0.298 |
| VF23 | −0.7241 | 1.5016 | [−3.6695, 2.1683] | 0.624 |
| VF24 | −2.4735 | 2.0545 | [−6.5204, 1.3948] | 0.229 |
| VF25 | −0.0420 | 2.6614 | [−5.2189, 5.5348] | 0.986 |
| VF26 | −1.9174 | 2.0246 | [−5.9303, 2.3030] | 0.345 |
| VF27 | −4.6170 | 2.7248 | [−9.7895, 1.1205] | 0.096 |
| VF28 | 4.5282 | 2.9070 | [−2.3001, 8.8359] | 0.202 |
| VF29 | −4.6475 | 2.6019 | [−9.7920, 0.3106] | 0.067 |

| Covariate | Estimate | Bootstrap SE | 95% CI | $p$-value |
|---|---|---|---|---|
| VF30 | −1.5992 | 1.8574 | [−5.5043, 1.6875] | 0.294 |
| VF31 | 0.7623 | 2.0298 | [−2.7546, 5.1378] | 0.606 |
| VF33 | −1.4154 | 1.2216 | [−3.9614, 0.6820] | 0.190 |
| VF34 | 0.9300 | 1.1479 | [−1.4252, 3.1472] | 0.466 |
| VF35 | −2.4346 | 1.5201 | [−5.1943, 0.7145] | 0.144 |
| VF36 | 0.7484 | 1.4691 | [−2.2744, 3.6347] | 0.676 |
| VF37 | −1.1999 | 2.1288 | [−5.5216, 2.9894] | 0.556 |
| VF38 | 1.5452 | 1.8560 | [−2.1044, 5.1512] | 0.386 |
| VF39 | −2.2188 | 1.7798 | [−5.7668, 1.3339] | 0.224 |
| VF40 | 0.8376 | 1.5046 | [−1.9421, 3.9402] | 0.549 |
| VF41 | −0.9182 | 1.6364 | [−4.0978, 2.4663] | 0.556 |
| VF42 | 0.6432 | 2.4218 | [−3.5035, 6.0623] | 0.706 |
| VF43 | 0.7816 | 2.1333 | [−2.8436, 5.5297] | 0.600 |
| VF44 | −2.1332 | 1.9133 | [−5.2422, 2.0390] | 0.341 |
| VF45 | 1.2080 | 1.5056 | [−1.4518, 4.4074] | 0.352 |
| VF46 | −2.2190 | 1.8511 | [−5.7743, 1.3990] | 0.222 |
| VF47 | 2.2628 | 2.2334 | [−1.8410, 7.0308] | 0.241 |
| VF48 | 1.2110 | 2.1274 | [−2.6759, 5.6408] | 0.504 |
| VF49 | −1.9245 | 1.7834 | [−5.6493, 1.4426] | 0.227 |
| VF50 | −0.2721 | 1.2640 | [−2.5242, 2.3640] | 0.928 |
| VF51 | −0.4740 | 1.0120 | [−2.6796, 1.2211] | 0.508 |
| VF52 | −1.3907 | 1.1003 | [−3.7335, 0.5389] | 0.167 |
| VF53 | 0.5845 | 1.5011 | [−2.2244, 3.7814] | 0.629 |
| VF54 | 1.6629 | 1.9435 | [−2.1976, 5.2682] | 0.457 |
| VF55 | 0.8503 | 2.2200 | [−3.6169, 5.1068] | 0.795 |
| VF56 | 0.4785 | 2.1697 | [−4.4390, 4.2848] | 0.926 |
| VF57 | 1.9185 | 1.8760 | [−2.3761, 4.8878] | 0.422 |
| VF58 | 0.8606 | 1.8375 | [−2.9659, 4.2252] | 0.752 |
| VF59 | 1.1116 | 1.8124 | [−2.2541, 4.8736] | 0.488 |
| **VF60** | **2.4809** | **1.3808** | **[0.0853, 5.3715]** | **0.042** |

**Empirical Findings.** Our VCTR model recognized a few patterns:

- The estimated coefficient functions $\hat{\alpha}(t)$ vary noticeably over age, suggesting that the association between ROI fundus images and IOP is age-dependent.
- In contrast, the estimated $\hat{\sigma}(t)$ changes only mildly over age, so the remaining noise variation does not appear to be strongly age-driven.
- The scalar-covariate analysis identifies three significant VF locations, VF00, VF04, and VF60.
- The estimated $\hat{\rho} = 0.617$ indicates moderate positive within-subject dependence between the two eyes, supporting the use of the paired-eye covariance structure rather than treating them iid.